Overview

Influenza A is a seasonal respiratory infection that typically breaks out during a specific period each year. Although each outbreak cycle is limited, seasonal influenza — of which influenza A is a major cause — imposes an extremely heavy global disease burden every season. According to the World Health Organization (WHO), seasonal influenza causes approximately 3–5 million cases of severe illness and 290,000–650,000 influenza-related respiratory deaths each year.[1] The epidemiology of influenza A shows a distinct cyclical pattern: after the peak of an epidemic, case numbers fall rapidly and public vigilance gradually fades; then, when the next season arrives, the mutated influenza A virus returns once again to threaten human health.

To improve the prevention and control of influenza A, we hope to use synthetic biology, with Saccharomyces cerevisiae as the chassis, to achieve capture and detection of influenza A virus. This is how the Echo Yeast project came into being. To build a device that is genuinely applicable and socially valuable, we cannot limit ourselves to theoretical reasoning and experimental validation inside the laboratory. We need to cooperate with people outside the lab, improve our project through the questions and suggestions of experts in various fields, and make our project fit the real circumstances and needs of society. We began by mapping out our project's stakeholders, including respiratory clinicians, frontline nurses, researchers and institutional staff. The suggestions they raised provided a realistic basis for the conception of our project, guided the optimization of our approach, offered reference points for model construction, and refined the equipment's application scenarios as well as many practical details of implementation. It is precisely with the assistance of experts from all walks of life that our project design has been able to align more closely with real-world usage needs.

Framework

To conduct rigorous, orderly and genuinely valuable social practice, a sound guiding framework is indispensable. With such a framework, we have a clear line of thinking when we need to carry out social practice, and we are able to make corresponding adjustments more orderly and conveniently during that practice.
We referred to the Action Research model and reviewed several of its classic formulations: the approach is commonly associated with Kurt Lewin, whose 1946 paper set out a spiral framework of research closely bound to practice, described as "planning-action-fact-finding";[2] afterwards, Susman and Evered refined it into five

phases in 1978,[3] and Kemmis and McTaggart further summarized it in 1988 as the cycle of "plan-act-observe-reflect".[4] Drawing on these formulations, we adopt the four-step cycle of "diagnose-plan-act-observe & reflect" as the practical framework of this project, whose intellectual origin is precisely Lewin's Action Research Spiral Model.

The defining characteristic of this cycle is that it is not a closed circle but takes the form of an upward spiral: the end of each cycle is the beginning of a new plan. It also embodies the unity of knowledge and action. We research through continuous social practice, and we engage in more social practice through that research, binding our project tightly to reality.

Where knowledge is genuine and earnest, that is already action; where action is clear-sighted and discerning, that is already knowledge. The effort of knowledge and action can never be separated.

— Wang Yangming, Instructions for Practical Living (Chuanxilu)[5]

Diagnose: Identify the problem and the motivation. What is the purpose of this social practice? What is the core question of this interview?

Plan: Design the plan for this social practice. Confirm who we will interview. What information might we obtain from them?

Act: Begin the interview, collect feedback, and record the interviewee's viewpoints and opinions.

Observe & Reflect: Analyze the feedback, revise and iterate the project design, and apply the feedback in practice. At the end of this round, begin identifying new problems and enter the next cycle.

Stakeholder

After establishing action research as our framework for social practice, we faced a key question: for Echo Yeast to truly serve public health, it cannot remain confined to technical reasoning in the laboratory. It must move into real scenarios and listen to voices from different fields. To this end, we systematically mapped out the project's stakeholders, hoping to obtain feedback on the project's positioning, technical route, application boundaries and path to implementation from multiple dimensions: clinical practice, disease control, scientific research, education, hardware engineering, mathematical modeling and peer exchange.

We first divided stakeholders into several categories. Clinical medicine and end-demand stakeholders help us understand the real pain points and usage scenarios of

influenza A detection. Public health and regulatory institutions examine the project's feasibility and boundaries from the perspective of disease control and compliance. Academic researchers and technical advisors provide professional guidance on experimental design, chassis engineering and molecular docking. Education, outreach and communication targets allow us to promote synthetic biology while collecting genuine feedback from the public and young learners. Hardware engineering and device translation experts help us judge airborne virus enrichment, component selection and competition delivery strategy. Mathematical modeling and computational simulation advisors provide methodological support for molecular docking, structural modeling and data analysis. Finally, the academic community and peer exchange also provide important support for the project's continued advancement.

These stakeholders are not a simple list; they correspond to different stages of the project, from technical validation to practical application. Through interviews we collected opinions from different perspectives, recorded them one by one, evaluated their feasibility, and translated them into concrete actions for project iteration. It is through these questions, suggestions and feedback that Echo Yeast has been able to continually calibrate its direction, moving step by step from a technical concept toward a problem-oriented, demand-driven biosensor design.

Stakeholder list

Clinical medicine and end-demand: Wenjie Wang, Kai Fu, Manqing Deng
Public health and regulation: experts in relevant fields
Academic research and technical advisors: Xinyuan Qiu, Zhouqing Luo, Hui Wu, Honghua Jia, Haili Zhang, Linsen Huang
Education, outreach and communication: Shenzhen First Vocational Technical School, outreach to incoming students
Hardware engineering and device translation: Rongkang Deng
Mathematical modeling and computational simulation advisors: Xiaolong Liu
Academic community and peer exchange: online exchange meeting on yeast chassis engineering, APiC, South China exchange meeting, Tsinghua Synthetic Biology Innovation Competition, SynBio Challenges

List of interviewees

Wenjie Wang, Director of the Medical Technology Department, a Grade-II hospital in Shenzhen
Kai Fu, MD, Sun Yat-sen University
Manqing Deng, pediatric clinician at a Grade-III hospital in Shenzhen
Rongkang Deng, manager at a Shenzhen biotechnology company

Xinyuan Qiu, expert in synthetic biology
Expert in the field of public health and disease control
Zhouqing Luo, Professor, Xiamen University
Staff member in the field of health administration
Hui Wu, Professor, Dalian University of Technology
Honghua Jia, Research Fellow, Nanjing Tech University
Haili Zhang, PhD, Academy of Mathematics and Systems Science, Chinese Academy of Sciences
Shenzhen First Vocational Technical School
Outreach to incoming students, 2026
Xiaolong Liu, Founder, Shenzhen Ankeming Technology Co., Ltd.
Linsen Huang, R&D Manager, Shenzhen Lianghe Biotechnology Co., Ltd.



Wenjie Wang

Director of the Medical Technology Department, a Grade-II hospital in Shenzhen

Diagnose:

The project's technical direction and its actual application positioning were still unclear, and the team lacked first-hand understanding of real influenza A detection scenarios.

Plan:

The team designed interview questions around mutation mechanisms, clinical testing, treatment windows, reagent updates, application scenarios and the feasibility of self-testing.

Act:

Interviewed Dr. Wenjie Wang to obtain his views on hospital testing logic, the use of specific antiviral drugs, kit updates and air-based detection.

Reflect:

We recognized the gap between laboratory models and real needs, and decided to focus on the conserved stem region of HA, testing in enclosed spaces, and proactive early-warning applications.

The team had by then largely settled on its technical route: using engineered yeast as the chassis, displaying nanobodies on the surface to recognize the conserved stem region of the influenza A virus HA protein, and building a biosensor capable of capturing airborne virus and outputting a detection signal. But settling on a technical route does not mean the project is viable. We still needed a frontline clinical laboratory expert to assess its practical applicability, and we wanted to speak directly with clinical laboratory specialists to understand the real pain points, treatment strategies and prevention needs of influenza A in actual medical settings.

Dr. Wenjie Wang pointed out that the existing IVD industry only takes effect after the virus has already broken out in the human body and reached the detection threshold, whereas our air-monitoring sensor could potentially shift the prevention and control logic from passive diagnosis and treatment to active prevention, issuing a warning before the virus enters the human body, which has important public health value. However, he also identified shortcomings in both the technical and application aspects of our project:

- Air detection involves many variables; virus concentrations in open environments are low, and the effective range of the sensor may be limited.

- The effective range and limiting conditions are not yet clear. Testing is needed on effective measurement range, environment size, and the stage at which the virus is present; otherwise it will be difficult to define the real application boundary.
- The risk of false positives and specificity problems is prominent. According to the expert's estimate, the false-positive rate of self-test reagents may reach 50%–70%; if the sensor moves toward portable self-testing, it must address specificity and threshold determination.
- Demand for clinical subtyping is limited. Hospitals mainly test whether the virus is positive or negative in order to guide treatment, and do not distinguish subtypes, since subtyping is released by the CDC.
- Implementation and standardization are insufficient. The project is still at the laboratory prototype stage; production, cost, standardization and actual deployment still need to be addressed.

Drawing on his frontline experience, the expert also advised us to focus on "detecting the virus in the air" rather than distinguishing subtypes; to prioritize testing the effective range in enclosed spaces, controlling the fixed variables first before studying the variable ones; to clearly define limiting conditions such as effective measurement range, environment size and virus stage; to refer to immunochromatographic methods when developing portable self-testing, but to strictly control false positives; and to proactively contact reagent manufacturers to promote translation of the results. In essence, these suggestions required us to fill four categories of data: sensitivity, specificity, environmental adaptability and application boundaries.

The team translated the expert's suggestions into the following. First, firmly choose the conserved HA stem region to achieve broad-spectrum detection and downplay subtype discrimination. Second, prioritize designing enclosed-space experiments and establish a variable-control scheme. Third, treat specificity and false-positive control as priorities for subsequent optimization. Fourth, clearly define the effective measurement range and environmental limiting conditions. Fifth, treat this interview as the point at which the project moved from a technical concept toward a problem-oriented approach, and it also made us keenly aware that for a cutting-edge sensor to truly serve public health, it needs not only innovation, but also the ability to withstand scrutiny from clinical scenarios, environmental variables and practical application.

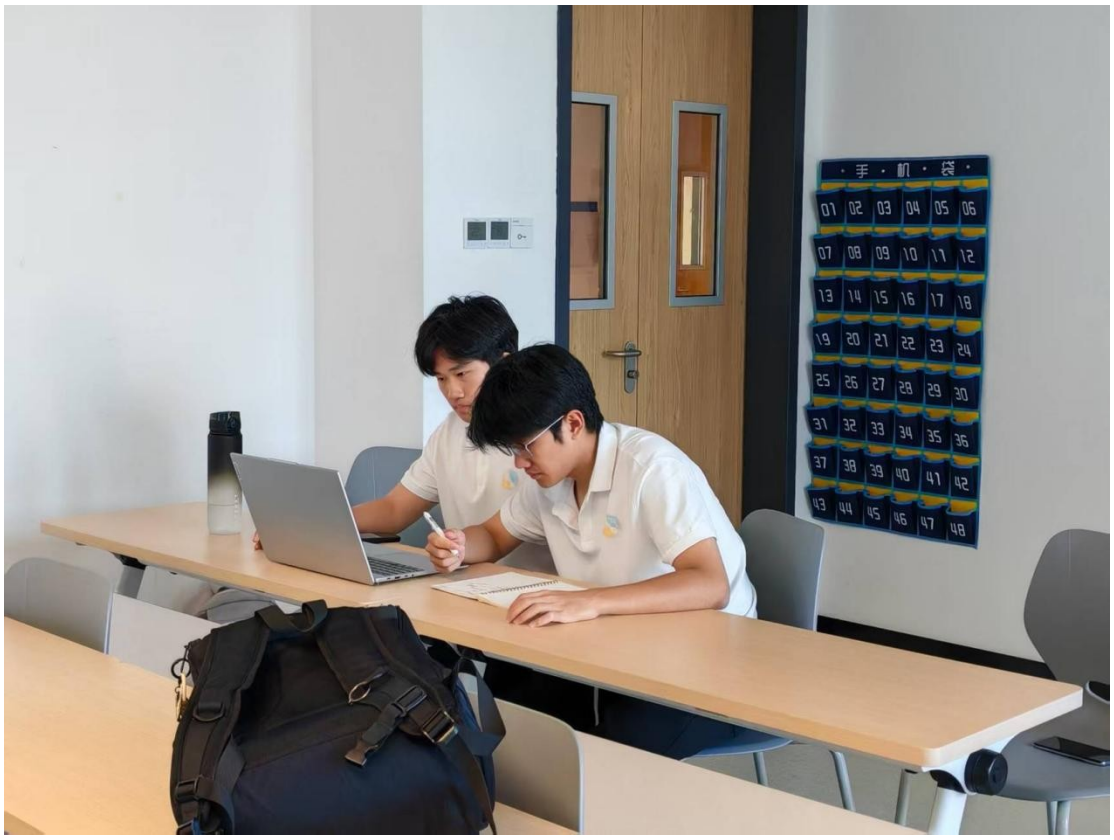


Figure 1. Photo of the offline discussion with expert Wenjie Wang

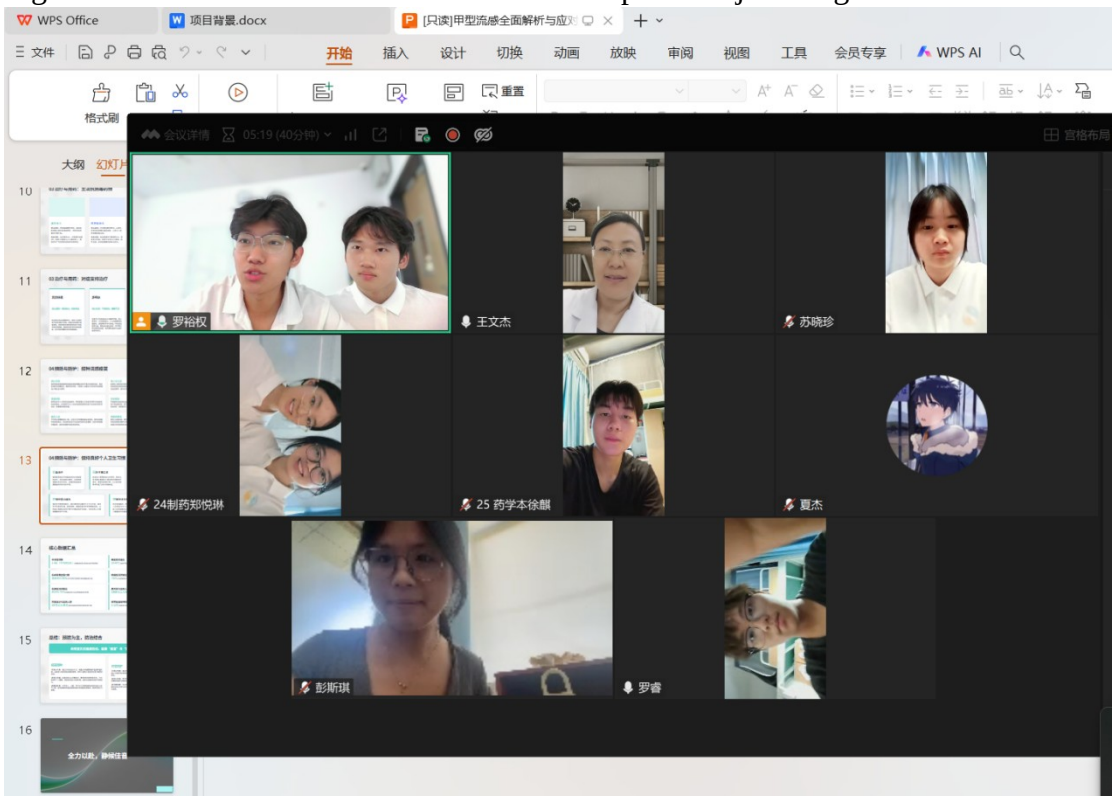


Figure 2. Photo of the meeting with expert Wenjie Wang



Kai Fu

MD, Sun Yat-sen University

Diagnose:

The environmental air monitoring direction was acknowledged, but questions remained regarding timeliness, virus concentration, cost and the boundaries of expansion.

Plan:

We designed questions around response speed, the reporter system, yeast viability, device cost, clinical scenarios and future expansion.

Act:

Interviewed Kai Fu, MD, Sun Yat-sen University, to obtain his views on clinical feasibility, breath-based detection and grassroots promotion.

Reflect:

Confirmed that a single-person breath mode is preferable; we need to shorten response time, control costs, and be cautious about expanding toward lung cancer detection.

The project had already focused on the conserved HA stem region, downplaying subtype discrimination, and had begun exploring the air-monitoring direction. But questions remained to be calibrated: the yeast sensor's response time, the reporter system, cost, the yeast's viable period, comparison with existing technologies, whether environmental air or human exhaled breath is preferable, and whether the system could be extended to influenza B or lung cancer.

During the interview, Dr. Kai Fu endorsed the concept of "using one organism to detect another organism," which he considered to have distinctive value in the field of synthetic biology. He specifically noted that people with respiratory infections or lung infections exhale virus particles carried out with aerosols, so the device would perform better on this group. But he also cautioned that, based on his experience, the yeast signaling pathway takes about 1–2 hours from receiving a signal to expressing fluorescence, and that if this time is too long, the device's timeliness will be poor; environmental airborne virus is diluted and influenza A can usually only persist in the air for a few hours, whereas virus inside the human body is more concentrated and therefore easier to detect.

Based on these judgments, Dr. Kai Fu's suggestions centered on four directions. First, make single-person breath testing the primary mode, supported by an enrichment

device, with environmental sampling as a supplement. Second, optimize the signaling pathway to shorten response time: fluorescence requires photoelectric conversion and software visualization, while blue/white screening can be judged by the naked eye, though its expression time remains to be determined. Third, control hardware cost: according to the expert's estimate, the cost of the active breath device could be brought down to around 10 yuan, which is what would make grassroots promotion feasible. Fourth, conduct cross-disciplinary interviews with experts in respiratory medicine, biomedical engineering and virology, and suspend the lung cancer detection idea for now, focusing on influenza A and other respiratory viruses; if a highly specific breath biomarker is found in the future, expansion can be reconsidered. He also reminded us to clarify the yeast's viable period and replacement scheme: based on his experience, activity peaks 2–3 days after resuscitation and can be maintained for 1–2 weeks.

Dr. Kai Fu advised us to focus subsequently on shortening detection response time and controlling hardware cost, and to prudently assess clinical fit when expanding the technology. This led the team to understand the possibility of moving from "environmental air monitoring" to "single-person breath detection." What we gained: the technical direction was endorsed, single-person breath testing is superior to environmental sampling, and low false positives, high sensitivity and ease of operation constitute differentiating advantages. What still needs to be supplemented: the expression time of blue/white screening, the cost of the complete device, the yeast's long-term viability and replacement scheme, quantitative data on specificity and sensitivity, a systematic comparison with PCR and test strips, and validation in grassroots applications. This interview advanced the project further on clinical feasibility, and reminded us that a cutting-edge technology must simultaneously answer four questions: how accurate, how fast, how expensive, and where it is used.



Figure 3. Photo of the meeting with expert Kai Fu

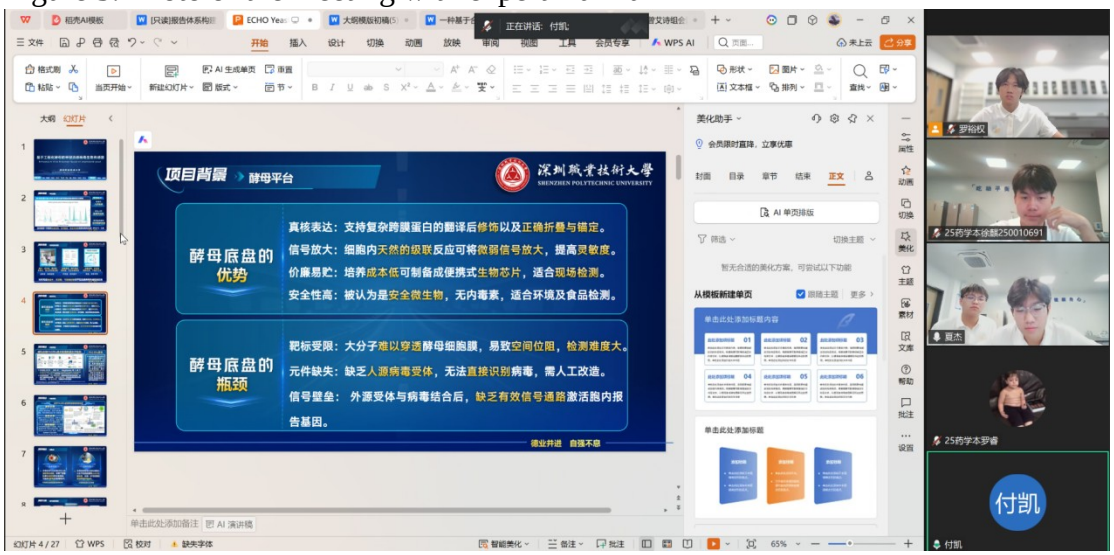


Figure 4. Photo of the discussion with expert Kai Fu



Manqing Deng

Pediatric clinician at a Grade-III hospital in Shenzhen

Diagnose:

The product is positioned as a detection reagent aimed at prevention, but there are still no answers at the implementation level regarding fit with susceptible populations, target audience, biosafety, clinical translation and pricing.

Plan:

Interview Manqing Deng, a pediatric clinician, to understand the cooperation of susceptible populations, parental acceptance, scenario selection, the path to market and ethical review.

Act:

Presented the project and gathered Ms. Deng's feedback on the current state of pediatrics, timing of medication, non-pharmaceutical prevention, technical route, market prospects, special populations and clinical translation.

Reflect:

Through this interview we further confirmed the project's positioning as a detection reagent aimed at prevention, and recognized that the B-end (institutional clients such as schools and government public health programs) should take priority over the C-end (household individual users); at the same time, low-load sensitivity, distinguishing live virus from fragments, cost control and compliance review are the keys to whether the project can be implemented.

We had gained a clear understanding of the technical direction, but that does not equal implementation of the project. We asked a pediatric clinician about several questions for which we had no answers at the time: children are a high-incidence group for influenza A, so can they cooperate with breath testing? What is parents' attitude toward testing and preventive medication? Do scenarios such as schools and kindergartens really welcome this? Should the product be classified as a detection reagent or a drug? Where does it sit in the clinical pathway: prevention, diagnosis or treatment? What processes must be cleared if it enters a hospital or a campus? What are the biosafety and ethical review requirements? Should the target audience prioritize families, schools, medical institutions or the government? At what price would it be competitive?

Ms. Deng endorsed the project's preventive positioning, considering it essentially a detection reagent rather than a drug, belonging to the "prevention" stage. If the virus can be identified at a low viral load stage, this is of great value for early isolation. Physical barriers (masks), frequent handwashing, school closures and influenza vaccines remain the most effective means of prevention. On biosafety, regarding the team's safety measures of auxotrophic culture medium plus filtration membranes plus a suicide system, Ms. Deng judged that as long as it is not sprayed into the human body and is used only for environmental/breath detection, the biological hazard is controllable.

On market and implementation, Ms. Deng advised prioritizing the B-end: schools are highly receptive, medical institutions are the most receptive, and the government can reduce medical insurance expenditure. On the C-end: households depend on price, and the product must be benchmarked against free vaccines and influenza drugs costing a few tens of yuan. She also suggested making the biological components into replaceable modules to expand into detection of other viruses, bacteria or antimicrobial resistance in mycoplasma. Clinical translation requires early engagement with GCP institutions and hospital equipment departments, striving to enter clinical guidelines and accumulating real-world epidemiological data; campus promotion must ensure safety and ease of operation, and for susceptible populations in particular there should be "zero operational threshold."

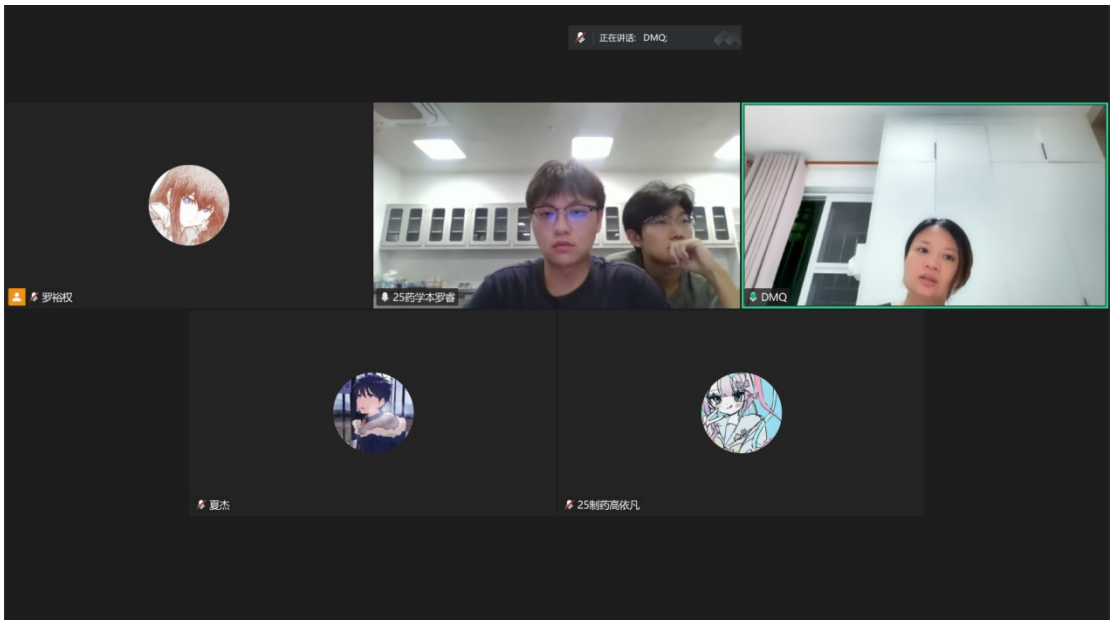


Figure 5. Photo of the meeting with Dr. Manqing Deng

Interview with an expert in the field of public health and disease control

Diagnose:
The project had already determined its technical route for influenza A detection, but regarding the rigor of topic selection, the choice of target genes, the bottleneck in air sampling, the registration pathway and methods of scientific validation, it still lacked judgment from the perspective of public health and disease control. The team wanted to test whether the approach of "first focusing on influenza A identification and then gradually expanding" could hold up, and asked the expert to point out the regulatory issues in technical design and subsequent implementation that are easily overlooked.

Plan:
We designed consultation questions around influenza subtyping, rigor of topic selection, detection technology design, air sampling, registration for implementation and research methodology, and exchanged views with an expert in the field of public health and disease control, hoping to examine the rigor of the project's topic selection and technical design from the perspective of a government agency for influenza surveillance and laboratory validation.

Act:
After the team presented the project, the expert first approached the issue from the rigor of topic selection, warning that detecting influenza A alone carries a risk of missed detection, since the clinical mainstream is dual testing for influenza A and influenza B, and the project presentation needs to face up to this shortcoming. On technical design, the expert advised that type diagnosis should preferentially use conserved genes such as M/NP/NS, that primers should be designed with reference to sequences from the past 3–5 years, and that conserved fragments of around 200 bp should be screened by BLAST; that antigen expression should preferentially use the conserved neck region of HA, and that a mosaic algorithm could be referenced to cover circulating strains. He also emphasized that a new project must be compared across multiple dimensions with mature methods such as rapid antigen tests and fluorescent PCR, rather than presenting only its advantages.

Regarding air virus detection, the expert affirmed its innovative value and recommended crowded places such as hospital emergency departments, subways and schools, but advised against entering the ICU. He pointed out that the core difficulty lies in air sample collection: virus particles are small, retention by filter membranes is limited, and collection schemes for bacteria and viruses are not interchangeable, so research should focus on collector selection, dry and wet collection methods, and membrane and concentration processes; if the front end cannot capture the virus, no matter how advanced the back-end detection is, it is useless. On sensitivity, airborne viral loads are generally low, which places extremely high demands on enrichment

and detection.

On implementation, the expert distinguished two paths, scientific research equipment and clinical medical devices: research use requires no stringent registration, but its results carry no clinical diagnostic validity; the expert advised that entering a hospital would require a Class III medical device registration certificate from the National Medical Products Administration, multicenter clinical trials and performance comparison with established reagents. On research methodology, the expert advised obtaining real clinical samples, processing data with statistical methods, objectively analyzing advantages and disadvantages, communicating more with industry experts, and not blindly following textbooks.

At the end of the interview, the expert and the team agreed on follow-up arrangements: the team would go back and improve its background research on influenza viruses and detection technologies; advance experimental validation with real samples and improve performance metrics and statistical analysis; subsequently engage experts from the medical device industry to discuss equipment design, evaluate the pros and cons of the scheme, and determine the R&D steps; and if questions arise later, the team could contact the disease control expert at any time, and the expert would also be willing to come to the school later to review the project.

Observe & Reflect:

From the perspective of public health and disease control, this interview provided the project with judgment criteria different from those of clinical end users and industrial translation. On topic selection, the team must face up to the risk of missed detection when testing only influenza A, and incorporate dual testing for influenza A and influenza B into subsequent planning. On technology, prioritizing conserved regions for target genes, choosing the HA neck for antigen expression, and referencing a mosaic algorithm provided clear guidance for molecular design. On air sampling, the expert identified front-end capture as the bottleneck of the chain, requiring focused research on collectors, filter membranes and concentration processes. On implementation, the registration requirements differ between the two paths of scientific research equipment and clinical medical devices, so the goal must be clarified early. On research methodology, real samples, statistical validation, cross-comparison and reproducibility became the basic principles for subsequent experiments and presentations. The expert also said he could be contacted at any time afterwards and was willing to come to the school to review the project, which provided important support for the project's continued advancement.

Interview with a staff member in the field of health administration

Diagnose:

The project already had preliminary directions on technical route, detection scenarios and product implementation strategy, but regarding industry standards for air virus monitoring, compliance classification of the product, biosafety regulatory requirements and the defense strategy for innovation and entrepreneurship competitions, it still lacked guidance from the perspective of health industry management. The team wanted to understand whether air virus monitoring has a business basis in the existing health system, whether there are official standards, and how the product should be positioned and made compliant.

Plan:
We designed consultation questions around the current state of the industry, scenario selection, experimental validation, biosafety, qualification and compliance, and competition defense, and exchanged views with a staff member in the field of health administration, asking for help in judging whether air virus monitoring currently meets the conditions for industry application, which category of management the product should fall under, and what norms experimental validation and biosafety need to follow.

Act:
After the team presented the project, the staff member first noted that the health system has no routine business of air virus detection in daily operations, and that China has no official standards or testing methods for air virus detection; in the interviewee's assessment, such equipment cannot serve as a basis for medical diagnosis and can only support environmental screening and risk warning, so it does not fall under medical devices. Existing air sampling mostly relies on laboratory equipment, and there is no mature commercial whole machine for online virus monitoring. Drawing on public health management experience, the staff member stressed that scenarios must not be over-generalized: open large spaces have low value, whereas high-value scenarios are fever clinics, elevators, high-speed trains, buses and enclosed classrooms, among which fever clinics are the most valuable and can be linked with disinfection equipment to form a closed management loop; the product could be considered as a sensing accessory for air disinfection machines, with points placed at both the air inlet and outlet to compare disinfection effects.

On experimental validation, the staff member advised combining laboratory simulation chambers with on-site testing, using pseudovirus and stained aerosols to simulate sneezing, varying distance, wind speed, temperature and humidity to test detection capability, and establishing reference risk concentration thresholds to answer "how much detected counts as high risk," while avoiding false alarms such as influenza B. On biosafety and compliance, the staff member reminded us to add UV-LED ultraviolet disinfection at the air outlet, to stop the machine automatically and sound an alarm if a membrane ruptures, to have membrane components self-disinfect on a schedule, and to add lysis buffer to the collection system to inactivate live virus; because the device would not perform clinical diagnosis, the interviewee advised that

integrating a disinfection module would place it under disinfection equipment, requiring disinfection-product filing; with no national standard yet, an enterprise standard could be drafted first. Regarding the competition defense, the staff member advised focusing on the detection principle, scientific logic and simulated experiment plan, with hardware as a future expansion direction; to make clear that the scenario pain point is risk warning in enclosed environments, not disease diagnosis; and to write the innovation points as environmental screening and warning for airborne viruses in enclosed environments, linkage with air disinfection equipment, and future expansion to multiple virus detection.

Observe & Reflect:
This interview helped the team draw clear boundaries: we only do risk warning in enclosed environments, not human diagnostic equipment; we lock onto core scenarios such as fever clinics, elevators, high-speed trains and enclosed classrooms; we focus on the route of a sensing accessory for air disinfection machines; and we design two sets of validation schemes, laboratory simulation and on-site testing, establish risk concentration thresholds, and write biosafety and compliance design into the project materials and the defense slides. The team recognized that air virus monitoring has neither mature standards nor commercial products, so the project should lock onto risk warning in enclosed environments, avoid the clinical diagnosis route, and treat biosafety and compliance design as a necessary part of the implementation plan.

Xinyuan Qiu

Our team uses Saccharomyces cerevisiae as the chassis and builds an airborne influenza A biosensor based on the PAGER system.[6] We have completed the construction of some genetic elements, but repeatedly encountered bottlenecks in experimental design, chassis engineering and iGEM review strategy. We therefore invited Xinyuan Qiu, an expert in synthetic biology, to guide our current experimental scheme and our iGEM matters. The expert identified the problems currently existing in our project, clarified the key validation that is presently missing, and offered valuable opinions.

Chassis and route: verify the premise first, then talk about simplification

The expert raised a key question that we had long overlooked: "Can an intact virus

pass through the cell wall to reach the cell membrane?" The expert pointed out that this is the physical premise for whether the device can work at all: the yeast cell wall is thick, and there is currently no evidence that intact virus particles can penetrate it and bind to the antibodies on hM1Dq; if validation fails, the chassis should be resolutely switched (for example to Xenopus oocytes) rather than continuing to invest in complex modifications. Based on this judgment, the expert's core advice was "the simpler the better": we had previously been stuck on problems such as knocking out endogenous interfering genes and validating the gating system, whereas the expert proposed a simplified receptor design modeled on RTK (receptor tyrosine kinase) architecture, replacing only the extracellular domain with an influenza A-binding protein and verifying downstream phosphorylation by Western blot. On the reporter system, the expert gave two paths: adjust the number of PREs or switch to a weaker promoter to reduce leakage; and molecular-level evidence must be added, using Western blot of MAPK phosphorylation to prove signal activation rather than relying only on indirect judgment from growth phenotype. The expert also emphasized that "strong antigen-antibody specificity" does not equal "high detection sensitivity": sensitivity must be defined by the number of virus particles detectable per unit volume, and our team precisely lacks the quantitative basis of "minimum detectable particles per unit volume." For multi-gene serial knockouts, the expert advised switching to Golden Gate cloning for plasmid construction to improve the success rate of cloning short sgRNA fragments.

Modeling: use it smartly, not exhaustively

We had hesitated over whether to build a full-chain ODE (ordinary differential equation) model from antigen to GFP, even though most parameters could only be obtained by fitting. The expert considered that approximating a multi-step reaction with a single Hill function is acceptable in review, but the parameters must come from experimental fitting or literature citations and must never be set arbitrarily; at the same time, the derivation of that approximation must be clearly written on the wiki. The real value of a model lies in first validating it by matching the curve to experimental data, and then using parameter sensitivity analysis to trace back the most sensitive step in order to guide experiments. As for our treating PAGER as a cascade of "antigen recognition to release of MT1 inhibition to GPCR activation" and attempting to encapsulate it with a three-state gate, the expert's judgment was that describing input-output and dynamic response with ODEs is more appropriate than three-state gate logic; mature algorithms already exist for sgRNA, so there is no need to expend effort on over-engineered forms such as Fourier transforms. The expert particularly reminded us that reviewers care about whether the model solves experimental problems, not whether the form is fancy; rather than stacking modules, it is better to give the model the ability to predict the entire "response-recovery" process.

HP and competition: necessity first, and the loop must be real

The expert pointed out that we should think hard about the "why" and build a complete project background and significance. He also noted that expert statements and social practice cannot replace factual evidence, and that "appeals to authority" should be avoided; we should use epidemiological and literature data to demonstrate the value of "airborne transmission detection" and the irreplaceability of the "yeast chassis." Human Practices is not judged by quantity or form, but by whether the practice truly influenced the project and drove improvement; for example, interview feedback at the hardware level must be translated into improvements in device design and verified. The expert's strategy for advancing was very clear: the immediate priority is to produce a functional receptor that can recognize influenza A antigen and to run through the first "simple and clear" DBTL (Design-Build-Test-Learn) cycle, and only then to iterate and update; he also advised studying the judging handbook and contributing to the community by submitting new parts or education projects.

Three shifts brought by the interview

(1) Necessity first: the epidemiological basis for airborne transmission detection, the irreplaceability of the yeast chassis, and the differentiating advantages over existing technologies are all essentially questions of "why the project exists," and all of them currently lack supporting data.
(2) Review and reflection: we once focused our energy on humanizing the GPCR, yet overlooked the premise of "whether the virus can pass through the cell wall"; simplified routes such as RTK that "only change the extracellular domain" deserve repeated self-scrutiny in subsequent decisions.
(3) Data is the only language of argument: the background cannot rest on verbal endorsement from experts, model parameters cannot be set from experience, and sensitivity must have a quantitative indicator. The expert also affirmed the value of the innovative idea of transplanting PAGER into yeast.

After the meeting we looked further into the literature and learned that Saccharomyces cerevisiae itself does not possess a classical receptor tyrosine kinase (RTK), so this route cannot be directly adopted in a yeast chassis. Nevertheless, the expert's idea of "simplifying at the level of receptor domains" reminded us to return to the judgment of "which step is truly necessary," and it remains very instructive for our trade-offs in receptor construction design.
After the interview, we raised the project's background research and chassis/receptor feasibility validation to the highest priority, and have continued to iterate and optimize the project on that basis.



Zhouqing Luo

Professor, School of Life Sciences, Xiamen University

Diagnose:

The project encountered bottlenecks in yeast sensor experiments, including difficulty in WB membrane protein characterization, uncertain membrane localization of the humanized GPCR, plasmid fluctuation in the reporter system, and doubts about cell wall penetration.

Plan:

Interview Mr. Zhouqing Luo of Xiamen University, consulting on membrane protein detection, receptor route, reporter system, cell wall permeability, signal amplification and response to viral mutation.

Act:

Presented the project's progress and problems, and gathered Mr. Luo's feedback on experimental operations, receptor modification, the reporter system, application scenarios and iteration directions.

Reflect:

Confirmed that we should prioritize modifying the yeast endogenous Ste2 receptor, first running the system through with antigen epitope short peptides; switch the reporter system to genomic integration; introduce intercellular communication for signal amplification; and focus on conserved epitopes, using a parallel multi-target design to cope with mutation.

Our project encountered considerable resistance in WB membrane protein characterization, membrane localization of protein expression, reporter system stability and the cell wall barrier; in particular, membrane protein detection produced many nonspecific bands and the internal control could not be resolved, and it was also uncertain whether the humanized GPCR could fold and localize correctly in yeast. We hoped to draw on Mr. Luo's experience in yeast engineering to identify the root of the problems and clarify the direction for optimization.

Drawing on his own experience in yeast membrane protein research, Mr. Luo first

helped us clarify the problems of WB characterization and the receptor route. Membrane proteins cannot be boiled at high temperature by conventional methods; he suggested gentle heating at 50–70 degrees Celsius for a short time, prioritizing mechanical disruption to extract protein, using dedicated membrane protein solubilization reagents, and introducing an HA tag to improve detection specificity and sensitivity. As for the receptor, he considered that the humanized GPCR is affected by glycosylation differences and carries high functional uncertainty, and preferred modifying the yeast endogenous Ste2 (alpha-factor receptor), designing it with protein structure so that it recognizes the target antigen; the signal peptide could also be replaced with a yeast surface display signal peptide to improve membrane localization.

Regarding the cell wall and the reporter system, Mr. Luo did not recommend the protoplast approach, because such cells are fragile and tedious to handle, and it would lose the core advantage of a yeast sensor; he suggested instead using yeast mutants defective in cell wall synthesis. He also pointed out that the HA protein — or the fragment relevant to detection — is only about thirty to forty kDa, so penetration is not the main obstacle, and validation at an early stage can directly use antigen epitope short peptides to run the system through first. On the reporter system, he advised changing the plasmid type to genomic integration to eliminate copy number fluctuation; mature modification schemes for MEL1 leakage have already been reported in the literature and can be found and referenced, so there is no need to blindly build a library.
On application scenarios and subsequent iteration, Mr. Luo stressed the need to explain clearly the unique advantages over PCR and colloidal gold, and to define the positioning of "direct sensing of intact virus and portable detection"; airborne virus concentrations are low, and if only a small amount of yeast is activated it is hard to detect, so he suggested using the yeast alpha-factor native cell communication pathway to build an intercellular communication plus positive-feedback signal amplification system. To address viral mutation, he advised a parallel multi-target strategy, co-expressing recognition elements for influenza A, influenza B and different mutant strains in the same chassis, prioritizing conserved antigen epitopes and avoiding high-frequency mutation epitopes.

This interview led the team to make clear adjustments in both experimental route and project positioning. Experimentally, we will turn to modifying the yeast endogenous Ste2 receptor, first running the system through with antigen epitope short peptides; the reporter system is planned for genomic integration; we will investigate intercellular communication positive-feedback signal amplification schemes; and membrane protein detection will switch to gentle heating and mechanical disruption, with evaluation of an HA tag. In terms of literature, we will consult materials on iScience membrane protein WB methods, Cell yeast GPCR modification, yeast surface display

signal peptides, cell wall defective mutants and MEL1 leakage modification. For project materials, the HP group will organize the materials and send them to Mr. Luo for review, and will sort out the presentation logic to distill the unique advantages of this technology compared with existing detection methods.



Figure 6. Photo of the meeting with expert Zhouqing Luo

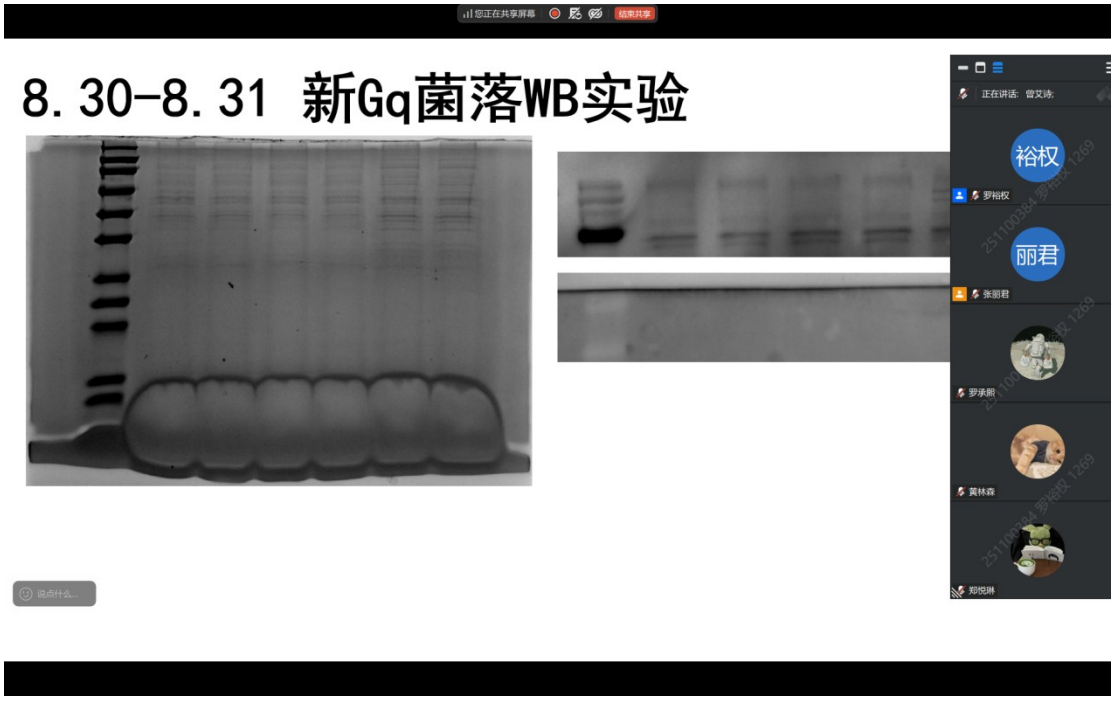


Figure 7. Photo of the discussion with Mr. Zhouqing Luo



Hui Wu

Professor, Dalian University of Technology

Diagnose:

The project encountered bottlenecks in all three modules of signal recognition, transduction and response: WB characterization was unsatisfactory, membrane protein expression was uncertain, and reporter system leakage was severe. Every node had problems.

Plan:

Interview Mr. Hui Wu of Dalian University of Technology, consulting on experimental progress, membrane protein expression, reporter system leakage, CRISPR knockout and iGEM strategy.

Act:

Presented the project's overall thinking and the progress of each module, and gathered Mr. Wu's feedback on experiment optimization, pathway completion, part submission and detection formats.

Reflect:

Confirmed that we should first complete one full pathway and achieve a visual response before optimizing components; membrane protein expression needs to reproduce the conditions reported in the literature, and reporter system leakage can be addressed by adding a repressor or changing the promoter.

As the project advanced, each module encountered resistance one after another: WB detection of membrane proteins gave unsatisfactory bands, the humanized receptor expressed poorly in yeast, the MEL1 reporter system leaked severely, and CRISPR knockout efficiency was unstable. We hoped that with Mr. Wu's guidance we could determine the root of the problems and clarify what should be prioritized at the current stage.

Mr. Wu first affirmed that the project's idea is novel and that using living cells to detect viruses has application potential. But he also pointed out directly that at present, from signal recognition to transduction to response, every node has problems

and there is considerable uncertainty, so he advised first reproducing conditions that have already succeeded in the literature and not innovating on every component. He emphasized that the immediate priority is to complete one full pathway and achieve a visual response, and only then consider replacing components to improve sensitivity and speed; only after a complete story is produced is there room for optimization later.
On specific experimental problems, Mr. Wu offered suggestions based on his experience. Membrane protein expression is inherently very difficult, and cross-species expression is even harder, so he advised first referring to conditions in the literature that can achieve active expression. On WB, he noted that yeast cell disruption cannot rely on snailase alone and suggested liquid nitrogen grinding, and that a low protein amount on SDS-PAGE is very likely an extraction problem. On reporter system leakage, the FUS1 promoter already has background expression without added alpha-factor, so one can try adding a repressor or screening other, more stringent promoters. On CRISPR knockout, he advised designing several targets and doing them at the same time to save the time spent on repeated attempts. Regarding iGEM part submission, he encouraged submitting as long as there is a slight trend, since a preliminary effect of the signal detection element is enough.
On application scenarios and detection formats, Mr. Wu advised comparing with environmental sampling RT-PCR and identifying the differences in time cost and sensitivity; yeast lyophilized powder can serve as the basis of a detection reagent, and gradient dilution during testing can establish a standard curve. He also mentioned that detection speed matters a great deal: RT-PCR takes about a day, whereas yeast culture may require two or three days, so we need to think about how to shorten the response time.

This interview led the team to readjust its priorities: rather than pursuing perfection in all modules at once, first complete one full pathway and achieve a visual response; experimentally, switch to liquid nitrogen grinding for cell disruption, optimize membrane protein extraction, try adding a repressor to the reporter system or changing the promoter, and run CRISPR with multiple targets in parallel; at the same time, organize components with preliminary trends for submission to iGEM and design standard curve experiments to evaluate the lyophilized powder detection format. Mr. Wu also affirmed the novelty of the project and encouraged us to first produce a complete story, and then gradually replace components to improve performance.



Figure 8. Photo of the discussion with expert Hui Wu

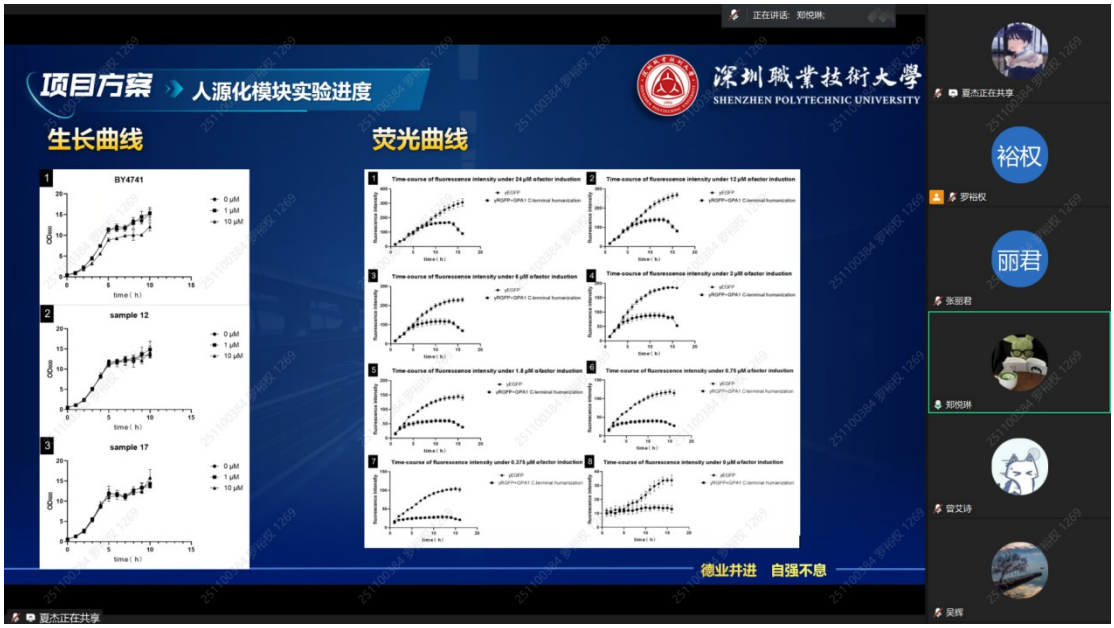
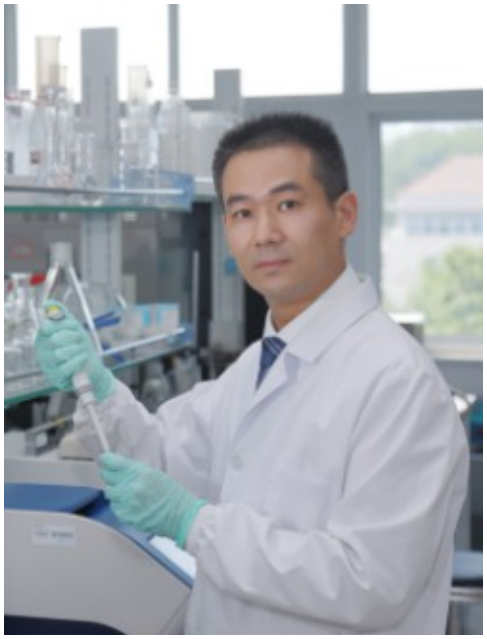


Figure 9. Photo of the meeting with expert Hui Wu



Honghua Jia

Research Fellow, College of Life Science and Pharmaceutical Engineering, Nanjing Tech University

Diagnose:

The project encountered resistance in protein expression, membrane localization, WB characterization and reporter system leakage, with the modules not yet fully working, and at the same time it needed to clarify how it differs from existing work.

Plan:

Interview Mr. Honghua Jia of Nanjing Tech University, examining the project design, effect indicators and validation methods from the perspective of synthetic biology and engineering.

Act:

Presented the project's thinking and the progress of each module, and gathered Mr. Jia's feedback on literature comparison, membrane protein validation, the reporter system, chassis selection and iGEM strategy.

Reflect:

Confirmed that we should focus on the final effects such as sensitivity and limit of detection, verify membrane localization using a fluorescent fusion, and reference existing yeast surface display work to clarify our novelty.

As the project has advanced to this point, construction has been completed, but functional expression and validation are still not fully working. We hoped to ask Mr. Jia, from the perspective of synthetic biology engineering, to help us judge the core problems of the current design and what should be validated first next.

Mr. Jia said that his own research direction is not yeast, and that on the whole he found the project's idea interesting, though the workload is considerable and some steps are not yet fully working. He particularly mentioned a paper by Na Wei's team at the University of Illinois, published in ACS Synthetic Biology in September 2024, which likewise uses a yeast biosensor to detect H1N1;[7] their approach is to display

the signal protein on the yeast surface, which is relatively simple and effective. He advised the team to consult that paper, compare the differences between surface display and membrane-bound schemes, and clarify our own novelty. He emphasized that iGEM values both innovation and the final effect; a biosensor must ultimately deliver on two key indicators, sensitivity and limit of detection/linearity, and effect is king.

This paper resonated with the question Mr. Xinyuan Qiu had raised earlier, "can the virus pass through the cell wall," and it also gave us a clearer understanding of why the surface display scheme works: it anchors the recognition element on the outer side of the cell wall, so the virus can be captured without entering the cell, thereby bypassing the premise of wall penetration on principle. From this we realized that surface display and membrane binding are not simply a matter of better or worse, but correspond to different design trade-offs and applicable scenarios, which gave us a basis for choosing between the two routes later.

On the uncertainty of membrane protein expression and localization, he advised fusing a fluorescent protein to the tail of the membrane component, first verifying whether it successfully reaches the membrane, and then observing the fluorescence localization with a confocal microscope; if membrane expression proves difficult, we could consider whether surface display might be more sensitive than full membrane attachment. Regarding the cell wall, Mr. Tang mentioned co-incubating with a low concentration of lyticase to hydrolyze some glycosidic bonds and enlarge the pore size for detecting bacteria, and Mr. Jia considered this idea usable for virus detection, making a very good combination of ideas.

As for reporter system leakage, Mr. Jia considered that the problem is not necessarily only in the promoter, since cascade regulation is fairly complex; changing to a weaker promoter or knocking out the chassis could both be tried, but the difficulty is high and he did not have a particularly good suggestion. The recognition element has little to do with the chassis; the key is whether it can be expressed correctly. Regarding iGEM review, he considered that the final result depends on the judges and that student performance also matters a great deal, and he recommended that the team talk with Ms. Guangna Liu (with experience leading multiple gold medal teams) and Mr. Wanlong Liu (who has worked on yeast and can be consulted about iGEM).

This interview moved the team from "burying ourselves in complex design" toward "validate first, then optimize, and look at the effect." We plan to consult the relevant 2024 ACS Synthetic Biology literature and compare surface display with membrane-bound schemes; use fluorescent protein fusion to verify whether the membrane protein successfully reaches the membrane and locate it by confocal imaging; evaluate the feasibility of treating the cell wall with lyticase; continue trying solutions to reporter system leakage; sort out effect indicators such as sensitivity, limit of detection and linearity; and contact Ms. Guangna Liu and Mr. Wanlong Liu for advice on iGEM and yeast experience. Mr. Jia also affirmed the ingenuity of the project's idea and the team's efforts, reminding us that in the end we must deliver a

demonstrable effect.

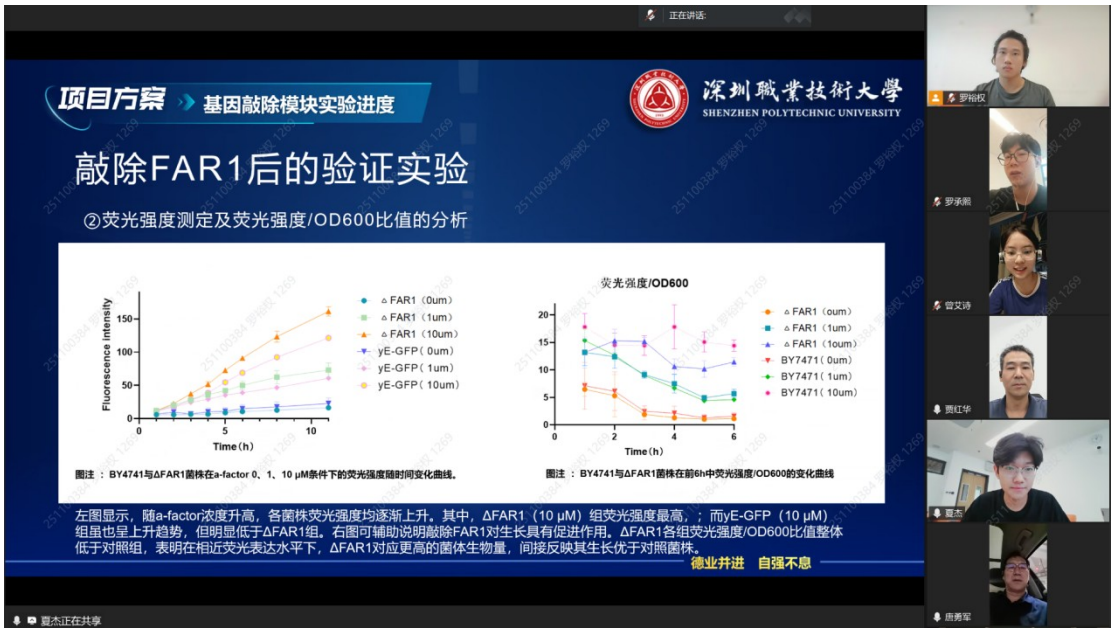


Figure 10. Photo of the meeting with Ms. Honghua Jia



Ms. Haili Zhang

PhD, Academy of Mathematics and Systems Science, Chinese Academy of Sciences

Diagnose:

The project's wet experiments had produced some data, but the modeling direction was still unclear, with multiple directions running in parallel, namely ODE, air diffusion and promoter screening, and with no priority; the existing growth curves and fluorescence time-series data showed visible differences to the naked eye, but lacked quantitative statistical support.

Plan:

Interview Ms. Haili Zhang of the Mathematics Teaching and Research Section of Shenzhen Polytechnic University, consulting on optional modeling modules, data statistics methods, bidirectional validation between modeling and experiments, and practical constraints.

Act:

Presented the project's principle, modular experimental strategy and four potential modeling directions, and gathered Ms. Zhang's feedback on the feasibility, priority and implementation path of each direction.

Reflect:

Confirmed that we should prioritize fluorescence time-series prediction and statistical hypothesis testing based on existing data, keep ODE as a fallback, and postpone the air diffusion model; modeling must serve the wet experiments and avoid excessive complexity.

The project already had some experimental data on humanization modification and signal reporting, including OD time series and fluorescence time series under an alpha-factor gradient, but the team was not clear how to distill from these data content that could support the iGEM modeling section. We also listed several directions, namely ODE, air enrichment, promoter screening and signaling pathway, but time was tight, and we needed the expert to help us judge which were worth doing and which

could be postponed.

Ms. Zhang evaluated, one by one, the four modeling directions the team had laid out, and gave a clear judgment of priorities. She considered that the ODE model depends on complete experimental data, that data at the present stage is insufficient, and that it is rather difficult to implement; that the air diffusion model is hard to validate with this project's wet experiments and is weakly coupled to the project, so it ranks low in priority; and that the promoter screening model can be referenced, but is not the most urgent direction at present.

What she most recommended was statistical analysis and time-series prediction based on the existing OD-fluorescence time-series data. On the one hand, hypothesis testing can be used to calculate P values, providing quantitative evidence for experimental differences discernible to the naked eye; on the other hand, the first and second derivatives of the fluorescence and OD time-series curves can be used to predict later trends from earlier data alone, shortening the detection time and moving closer to real-time detection. She pointed out that this direction can feed back into the experiments and aligns with the iGEM modeling scoring criteria. Regarding the humanization modification, she considered that at present it can only be assessed indirectly through growth curves, and statistical testing could be introduced; experiments to directly measure G protein binding efficiency completely have not yet been completed.

Taking into account the project's current time and data conditions, Ms. Zhang also reminded us that the wet experiments for some modules have not yet produced successful data, and that modeling depends heavily on measured data; time is tight, so it is not advisable to open up several complex modeling efforts at once; and modeling need not pursue excessive complexity, since its core is to serve the experiments and solve real problems rather than to look sophisticated in form.

This interview helped the team clarify the priorities for modeling: the highest priority is derivative-based time-series prediction using the existing OD-fluorescence time-series data, exploring the feasibility of using only earlier data to predict later results and shorten the detection time; the second priority is to add statistical hypothesis testing with P values for the humanization modification and the control experiments; the ODE system of differential equations remains a fallback, reusing literature parameters first and attempting fitting once more data is produced; and the air enrichment and viral spatial diffusion models can be postponed if time is tight. Going forward the team will continue collecting OD time series, fluorescence time series and growth curve data for each humanization modification group under an alpha-factor gradient, retaining biological replicates; complete comparative validation of single-module, double-module and full-pathway modified strains; confirm with Ms. Lijun Zhang whether statistically significant P values need to be reported; and after producing preliminary results from the time-series prediction model, arrange a second

interview with Ms. Haili Zhang to review the statistical methods, model logic and fitting scheme. Ms. Zhang also reminded us that modeling need not pursue sophisticated forms; what matters is that it can serve the experiments and solve real problems.

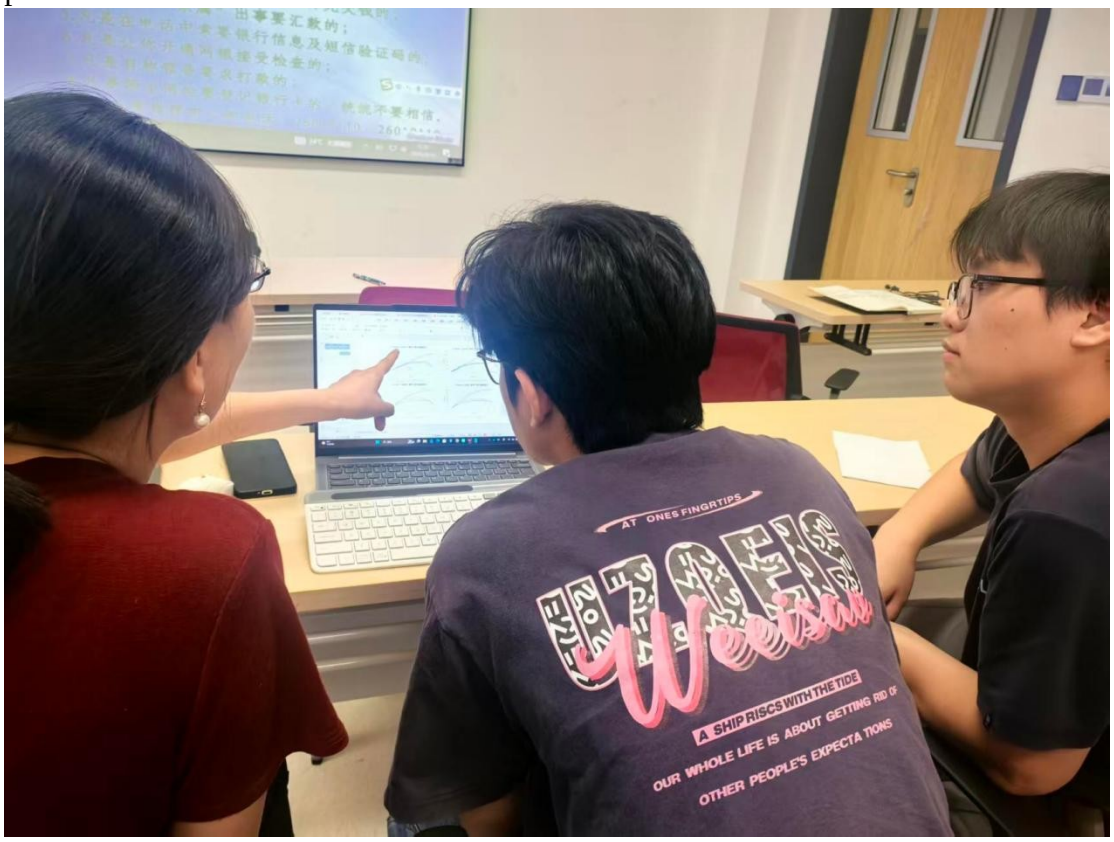


Figure 11. Photo of the discussion with Ms. Haili Zhang



Figure 12. Photo of the meeting with Ms. Haili Zhang



Xiaolong Liu

Diagnose:

The team had completed structural prediction of 16 nanobodies and obtained H1N1/H2N2/H3N2 HA antigens, and had performed preliminary docking with an online website, but the applicability of the tools, the quality of the modeling, the structural preprocessing and the depth of the results analysis had all not been verified.

Plan:

Interview Mr. Xiaolong Liu, consulting on the choice of molecular docking tools, modeling methods, presentation of results and the validation workflow.

Act:

Presented the current docking progress and the way the results were analyzed, and gathered Mr. Liu's feedback on tool algorithms, modeling tools, free energy decomposition, MD simulation and wet-experiment validation.

Reflect:

Confirmed that the current methodology has systematic shortcomings: we need to remodel with AlphaFold 3, verify the applicable scope of the docking tools, add MM/GBSA decomposition and MD simulation, and finally close the loop between computation and experiment through SPR/MST.

At the current stage, the team is using molecular docking to provide computational simulation support for the project, and has completed structural prediction of 16 nanobodies and the acquisition of three subtype HA antigens, with preliminary docking performed on an online website. But we are not sure whether these tools are applicable to a nanobody-protein system, nor whether the results analysis is professional enough and can withstand follow-up questions from reviewers. We therefore asked Mr. Liu to review our methodology and validation workflow.

Mr. Liu first pointed out that the online docking website the team is currently using may not have been designed specifically for protein-protein or antibody-antigen docking, and that the applicable scope of its algorithm needs to be verified;

nanobodies have a special structure, different from Y-shaped IgG, so general protein docking algorithms may not apply. He stressed that just because a website can produce results does not mean the methodology supports our system: we must go to the original literature and confirm whether the sampling algorithm covers protein-protein docking and whether it applies to nanobodies.

On modeling and structural preprocessing, homology modeling tools such as Swiss-Model give unreliable results when template homology is below 25%, so he advised switching to AlphaFold 3,[8] which he considered more reliable for this case and which provides confidence scores; after modeling, he recommended a Ramachandran plot assessment, with a project threshold of more than 95% of residues in allowed regions; and HA structures downloaded from the PDB need water, metal ions and salt ions removed, with hydrogens added and charges balanced.

On results analysis and validation, presenting only the total binding energy is not enough: MM/GBSA free energy decomposition should be performed, analyzing the contribution residue by residue to verify whether the key interactions are concentrated in the CDR region; for visualization, heat maps and PyMOL are recommended to show the binding details. Docking results represent only a rigid or semi-flexible state, so molecular dynamics simulation of at least 100 ns should be added to verify the stability of the complex; finally, actual affinity should be validated by SPR or MST wet experiments, forming a complete loop of "modeling-docking-analysis-MD-wet experiment." On tools, industrial-grade software such as Rosetta and Boltz are authoritative but have a high barrier to deployment; at this stage we can continue using online tools, but must confirm that the algorithm is public, has original literature and is widely cited by peers; PLIP can be used for interaction analysis. AI-assisted computation is acceptable, but the method must come from a path that is already open-source and peer-validated.

This interview made the team realize that molecular docking is not complete simply because "results come out"; the methodology itself must stand up to scrutiny. Going forward we will remodel the 16 nanobodies with AlphaFold 3 and add Ramachandran plot assessment; preprocess the HA protein by removing ions, adding hydrogens and balancing charges; verify the original literature and applicable scope of the docking website we used; perform MM/GBSA free energy decomposition and produce residue contribution heat maps; add molecular dynamics simulation to verify the stability of the complex; and finally close the loop through SPR/MST wet experiments. On the Wiki and in the report we will cite the original literature of the methods used, provide model assessment data, and clarify the applicable scope of the tools and the reasons for choosing them. In addition, Mr. Xiaolong Liu will continue to guide our mathematical modeling work afterwards, helping us steadily improve the connection between modeling ideas and experimental data.

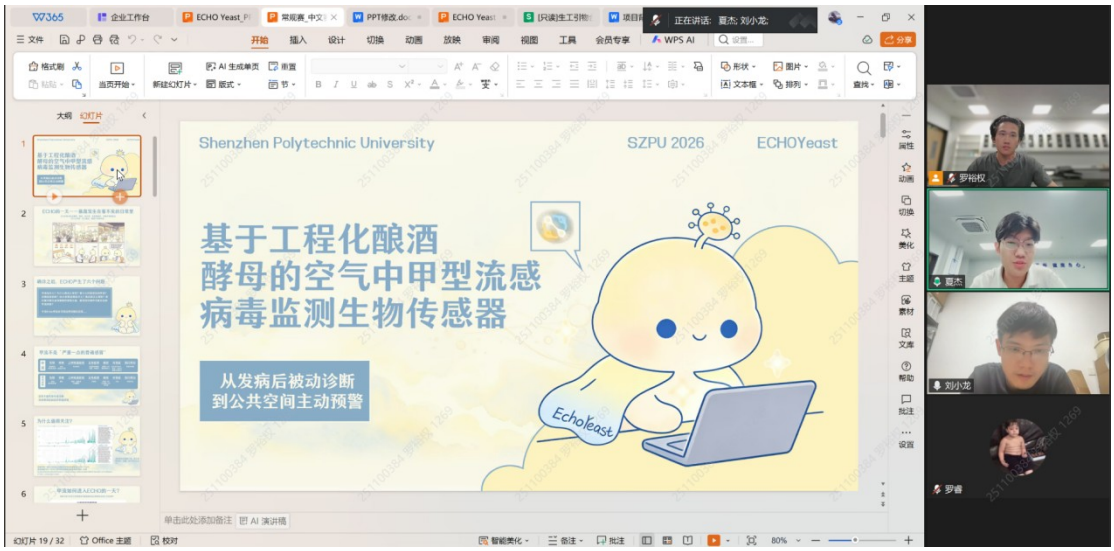


Figure 13. Photo of the meeting with expert Xiaolong Liu



Figure 14. Group photo of the meeting with expert Xiaolong Liu

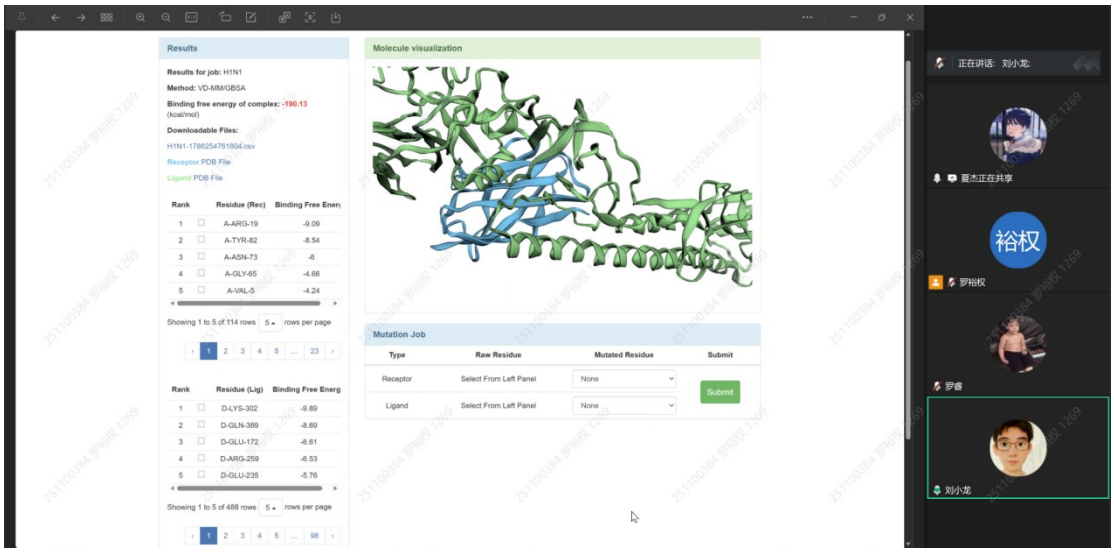


Figure 15. Photo of the discussion with expert Xiaolong Liu

Rongkang Deng

Diagnose:

The project's equipment concept was already clear, but the core bottleneck of airborne virus enrichment remained unsolved, and there was no experience in hardware engineering, R&D workflow, component selection or competition delivery strategy.

Plan:

Interview Mr. Deng, an engineer in medical device hardware development, consulting on equipment design, enrichment technology, R&D workflow, intellectual property and competition delivery.

Act:

Presented the equipment concept and working mode, and gathered Mr. Deng's feedback on enrichment ideas, R&D workflow, competition strategy and industry practice.

Reflect:

Confirmed that airborne virus enrichment is the greatest difficulty, and that at the competition stage we should prioritize completing the principle block diagram and a partial demonstrator rather than pursuing a complete prototype; the defense must show a thorough grasp of the principle to avoid being judged as falsification.

The project plans to develop equipment that can collect influenza A virus from ambient air, enrich it and then complete detection to provide a risk warning, targeting enclosed spaces such as schools and cruise ships. But we lack experience in hardware engineering, and in particular are unclear about airborne virus enrichment, component selection, R&D workflow and competition delivery, so we asked Mr. Deng to judge from the perspective of medical device development which parts are realistically feasible and how far we should go at the competition stage.

Mr. Deng first helped us sort out our thinking: the detection chemistry and optics are in fact manageable in difficulty, and the real bottleneck is capturing the virus from the air and enriching it in the liquid phase. Virus particles are too small, and if air is bubbled directly into the reaction liquid, most of it escapes as bubbles and dissolves with difficulty, while the liquid also evaporates after prolonged aeration. He proposed two schemes: first, vortex stirring, so that air and liquid make full swirling contact; second, membrane filtration, with multiple filter layers, the upper layer retaining

bacteria and the lower nanomembrane retaining virus, which is then washed off with eluent. The trouble with membrane filtration is that virus easily passes through the membrane and that higher pressure easily causes clogging, which can be alleviated with pulsed pressure or vibration. He also mentioned that enrichment of urinary exosomes follows similar logic: it can be done in the laboratory, but engineering it runs into practical problems such as time, clogging and impurities.

On timing, Mr. Deng felt that building a complete prototype in two or three months is unrealistic; he advised first producing the appearance drawing, the working principle block diagram and the module-level schematic diagrams to make the logic clear, and that a partial core-function demonstrator could also be made, without needing the whole machine to work end to end. But he particularly reminded us that the team must thoroughly master the whole scheme, and cannot hold up drawings while being unable to explain the principle, since that would be treated as falsification.

He also laid out a fairly clear line for the R&D workflow: first write the "Product Technical Requirements," setting out clearly everything that can be quantified, such as sampling duration, air pump flow rate, liquid volume, temperature and humidity, false-positive/false-negative indicators and validation methods, while also consulting industry standards and similar devices; then determine the technical route, first building a simple simulation device to verify whether the virus can enter the liquid phase and what the enrichment efficiency is, without rushing to a whole machine; then split hardware and software, with the lower computer handling the air pump, solenoid valves and temperature control, and the upper computer handling human-machine interaction, defining the communication protocol well; then build a prototype and iterate testing to find the balance between performance and cost. Since the equipment involves live virus, internal inactivation schemes and safety validation must also be considered in advance.

On tools and intellectual property, he advised learning programming in a targeted way around subtasks rather than studying broadly; AI can be used, but key parameters and regulations must be checked against the original materials. On innovation, the vast majority of products are not original from scratch: one can refer to air purification equipment and the enrichment devices in published papers, adjusting parameters to adapt on the existing basis and prioritizing the reuse of mature modules. On patents, the student stage is relatively lenient, but once commercialized, searches and design-arounds become necessary, and one cannot directly copy or reverse-engineer competitors' products. As for using yeast to recognize the HA protein, he said he had not researched similar commercial products; fragment-level research may exist in laboratories but complete devices are rare, and we would need to check the literature ourselves.

Finally he advised us to first produce the product technical requirements document, and then use it as the basis for drawing the appearance, structure and principle block diagrams as competition materials, with further drawing and refinement of the scheme to be discussed later.

This interview made the team realize that the main difficulty in building a complete piece of equipment lies in airborne virus enrichment. Going forward we will prioritize completing the product technical requirements document and clarifying the indicator requirements of our equipment; based on the technical requirements, produce the appearance drawing, working principle block diagram and module-level schematic diagrams as competition materials; and at the same time make a partial core-function demonstrator. On component selection, we will prioritize reusing mature submodules, referring to air purification equipment and published enrichment devices and adapting them rather than developing everything in-house; on patents, we will do proper searching and design-arounds. For defense preparation, team members will thoroughly master the logic of the whole scheme to ensure they can answer the judges' questions. Mr. Deng also said that drawing and refining the scheme could be discussed and coordinated afterwards, providing sustained support for the project's hardware progress.



Figure 16. Photo of the meeting with expert Rongkang Deng



Figure 17. Photo of the discussion with expert Rongkang Deng



Linsen Huang

R&D Manager, Shenzhen Lianghe Biotechnology Co., Ltd.; team adviser and experimental technical mentor

Diagnose:
Throughout long-term experiments in yeast chassis engineering and molecular cloning, the project needed sustained technical support to resolve specific bottlenecks in daily operations and scheme design.

Plan:
The team treated Linsen Huang as a long-term experimental technical mentor, regularly summarizing operational difficulties and design questions encountered during the project and discussing them with him to seek targeted optimization advice.

Act:
Drawing on extensive experience in yeast manipulation, Linsen Huang provided detailed experimental guidance and optimization ideas on specific steps including colony PCR lysis, reporter system leakage, vector construction and transformation.

Reflect:
This sustained guidance helped the team resolve many critical experimental bottlenecks, and members came to appreciate that solid, standardized experimental technique is an important foundation for the smooth progress of synthetic biology engineering design.

At every stage of the project, whenever the team encountered bottlenecks in experimental operations or scheme design during yeast chassis engineering and molecular cloning, it consulted the team's adviser and experimental technical mentor, Linsen Huang. This interaction ran throughout the project's long experimental phase rather than being a single interview. From low colony PCR identification efficiency to reporter system leakage and difficulty in screening transformants, Linsen Huang drew

on extensive yeast experience to provide practical optimization plans.

For rapid lysis in yeast colony PCR, Linsen Huang provided a simple heat-lysis method: pick a single colony after transformation into sterile water, spot about 2.5 μL onto a plate to preserve and number the strain, then mix the remaining suspension and place it in a PCR instrument for heat lysis using a program of 98 °C for 3 min and 4 °C for 2 min, repeated for 5 cycles; the lysate can then be used directly as a PCR template. For background leakage in the MEL1 blue-white screening system, he suggested designing degenerate primers to mutate the promoter region, constructing a mutant library and then screening it to obtain a more stringent promoter. If a transformation plate has too many colonies to pick single clones, he suggested washing the colonies off with sterile water, mixing and then plating a gradient dilution to improve screening efficiency.

In addition, for vector construction and transformation, Linsen Huang provided long-term, detailed guidance on key parameters including optimization of the PCR program for linearized vectors, redesign of the linearization strategy, preparation and recovery of PCR reactions, exploration of colony PCR conditions, transformation conditions for homologous recombination templates, and the ratio of insert to vector fragment added during yeast transformation. Through this continuing exchange, the team not only solved specific operational problems but also substantially improved the success rate of molecular cloning and yeast transformation, establishing a solid technical foundation for subsequently completing the full signaling pathway.

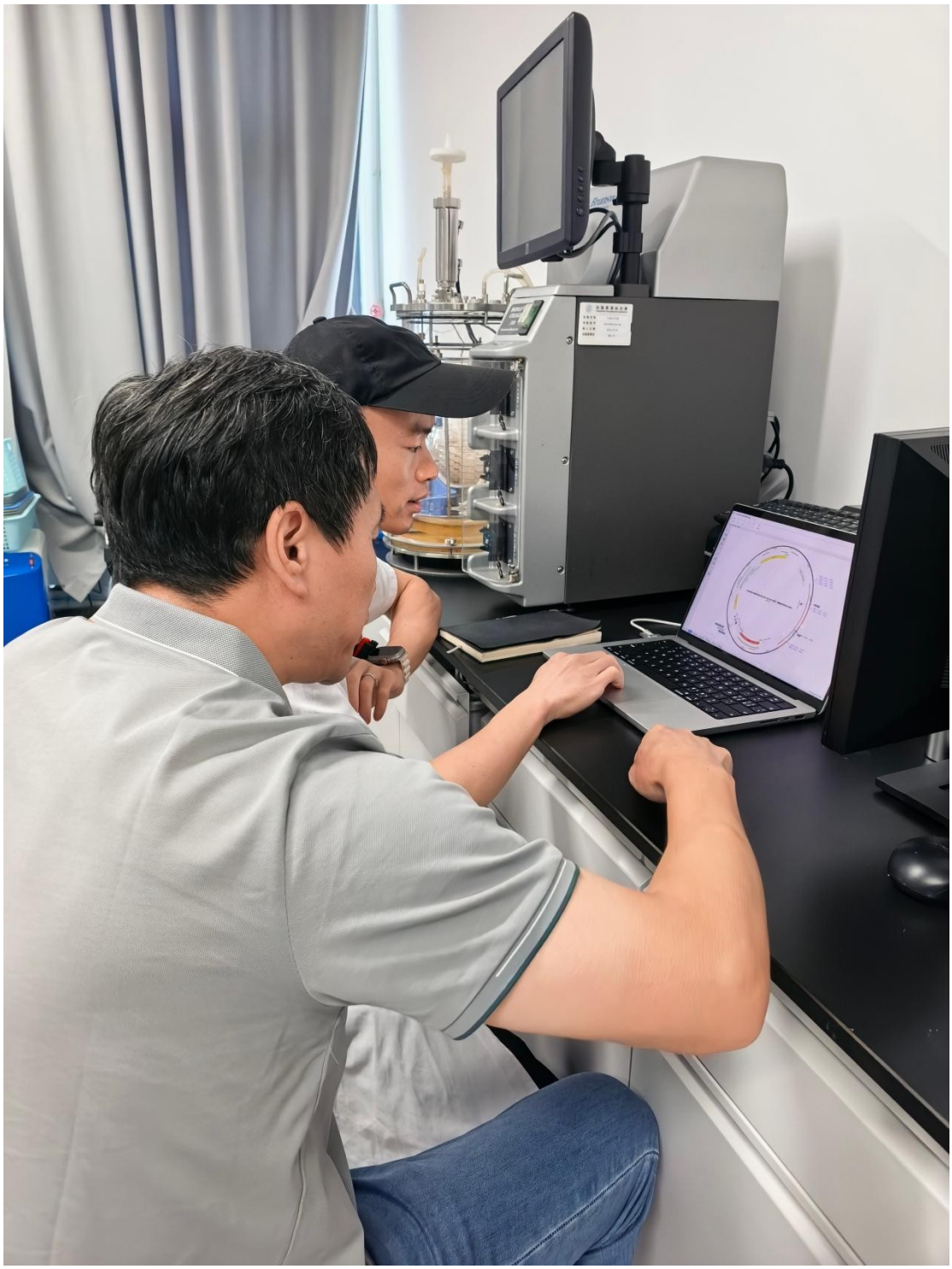


Figure 18. Discussion with team adviser Linsen Huang

Academic exchange

In the course of the project we gained guidance in clinical practice, disease control, industry and modeling through expert interviews, and also engaged actively with the wider synthetic biology community. Our activities ranged from high-level competitions — the First Tsinghua University Synthetic Biology Innovation Competition and the offline final of SynBio Challenges (SC) — to regional events including the South China iGEM team exchange, the Asia-Pacific iGEM team exchange (APiC) and an online meeting on yeast chassis engineering. Through these platforms we shared our project, discussed difficulties and learned from the experience of others, continually recalibrating our direction in response to peer feedback. These exchanges and competitions not only showed us what other teams had achieved but also helped us confront our own shortcomings; discussing each other's difficulties gave us a great deal of inspiration as well.

Tsinghua University Synthetic Biology Innovation Competition

At the First Tsinghua University Synthetic Biology Innovation Competition, the team presented Echo Yeast alongside teams from 44 universities, including Tsinghua University, Peking University, Tongji University and the Southern University of Science and Technology. Hosted by Tsinghua University, the competition aims to break down disciplinary, experimental-resource and prior-achievement barriers, creating China's first synthetic biology innovation platform centered on ideas and open to all.

At the venue, the team presented its engineered-yeast influenza A biosensor to judges and peers through a poster and an on-site explanation. Addressing the seasonal outbreak of influenza, widespread misconceptions about prevention and control, and the shortage of simple detection devices suitable for routine pre-screening, the team focused on the design logic of the project: using the PAGER system to overcome the difficulty that Saccharomyces cerevisiae has in recognizing the extracellular HA protein, engineering the yeast to recognize influenza A HA antigen and to report viral load quantitatively through a green fluorescent signal. The team also showed peers the planned format of the final detection consumable — a lyophilized yeast preparation that requires no cold chain, is inexpensive and does not depend on PCR or large instruments, making it well suited to routine environmental screening before each influenza season.

Through mutual visits and peer review with other teams, members not only introduced the project but also received candid feedback on implementation and scenario definition. As they reflected afterwards, the greatest gain was not the trophy but the exchange with fellow students: discussing and learning together generated many small ideas about the field, and explaining the project to peers deepened their

own understanding and exposed blind spots they had not previously noticed. The experience also pushed the team beyond a laboratory perspective. As team leader Jie Xia noted in his summary, the team realized that it needed to think more systematically about where the project would be used and what social value it would create; next steps include examining the product's practical value, investigating real user needs and refining the implementation plan.

In the end, for its novel concept, solid scientific foundation and strong on-site presentation, Echo Yeast received the competition's “Most Popular Award”. The competition recognized the team's progress to date and, by pushing members outside their comfort zone to answer questions without notes, strengthened their communication and adaptability — experience that will support the project's continued iteration.



Figure 19. Photo with teachers at the competition venue



Figure 20. Group photo at the competition venue

SynBio Challenges (SC) Offline Final

In August 2026, the team took part in the offline final of SynBio Challenges (SC) in Shenzhen. The event is listed among the implementation initiatives of UNESCO's International Decade of Sciences for Sustainable Development (2024–2033). It is hosted by the Chinese Society of Biotechnology and co-organized by Shenzhen University of Advanced Technology and the Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences, among others, under the theme “understanding through creation, application through creation”. According to the organizers, 512 teams from 9 countries and 108 universities entered this year's SC.

At SC, the team presented Echo Yeast in full. The project introduces the PAGER platform into yeast: nanobodies displayed at the yeast cell surface capture influenza A surface antigens, forming a two-key AND gate of “antigen + ligand (DCZ)” that signals through a Gpa1/Gαq chimera into the yeast pheromone pathway and drives FUS1-yEGFP or FUS1-LacZ readout. In the defense, the team explained the system's positioning for rapid detection of protein antigens such as influenza, its safety considerations (a GRAS chassis with additional containment), low cost and lyophilizable portability, and reported interim progress on expression of key fusion proteins and construction of the dual reporter system.

During the exchange, the team took part in mutual visits and peer review with teams from other universities, discussing data evidence, scenario focus and translation pathways, and receiving valuable suggestions from peers and judges. The team also

joined a roundtable dialogue between life science and life art, re-examining how it presents and communicates the project from an interdisciplinary perspective. In the formal defense, the team opened with a story, explaining the burden of influenza and why the yeast chassis is indispensable. Asked by the judges about the system's minimum detectable concentration in real air and its specificity, the team responded by drawing on background research, social practice and experimental progress, and recorded the quantitative data and validation plans that still need to be supplied.

Echo Yeast received a Gold Award in the SC awards. The team will turn the judges' and peers' suggestions into continued iteration: completing system calibration, improving receptor coupling efficiency and nanobody affinity, advancing scenario pilots with partner organizations, evaluating the stability of the lyophilized formulation in real samples, and continuing to refine safety, ethical and regulatory pathways.



Figure 21. Group photo of the full team at the competition



Figure 22. Award certificate



Figure 23. Photo of the team at the SC defense

Summary of the South China iGEM exchange meeting

At this South China iGEM exchange meeting, the team had face-to-face exchanges with several iGEM teams from South China, gaining a great deal from project sharing, academic discussion and mock review.

At the exchange meeting, teams from various places could share their projects. The team presented the influenza A air detection project on the big screen, explaining the overall design thinking from the yeast chassis and signal recognition to the reporter system. Other teams raised questions about the low concentration of airborne virus, yeast sensitivity, reporter system leakage, and the differences from traditional detection methods. The mock review session that followed let us experience the rhythm of an iGEM defense in advance; some answers were not thorough enough, but precisely for that reason they exposed the weak points in our project presentation. We recorded these questions and feedback one by one, and organized them into a list after the meeting, incorporating them into the directions for subsequent improvement. The questions collected mainly fell into four categories: at the experimental level, yeast response time, signaling pathway validation and virus concentration thresholds; at the mechanical and cost level, model calculation, cost control and yeast replacement; at the application scenario level, equipment installation location, data coverage in large scenarios and the feasibility of household self-testing; and in other respects, the advantages over existing detection technologies, the project's actual role, whether similar research exists, and biosafety approval. These questions essentially cover the key links of the project from experiment to implementation.

Beyond the formal exchanges, the team visited other teams' project displays to learn about their design thinking and experimental arrangements in directions such as environmental detection, treatment and materials. Different teams had their own distinctive approaches to modeling, hardware and human practices, giving us a good deal of reference. The games and interactive sessions at the meeting allowed the teams to communicate their progress and difficulties in a relaxed atmosphere, and some experience that would not normally be written into the wiki was shared in these exchanges.

This exchange meeting showed us our own gaps and clarified the direction for improvement. The team will put the suggestions received into practice in experiments and presentations, continue to refine the project, and prepare for the follow-up work.

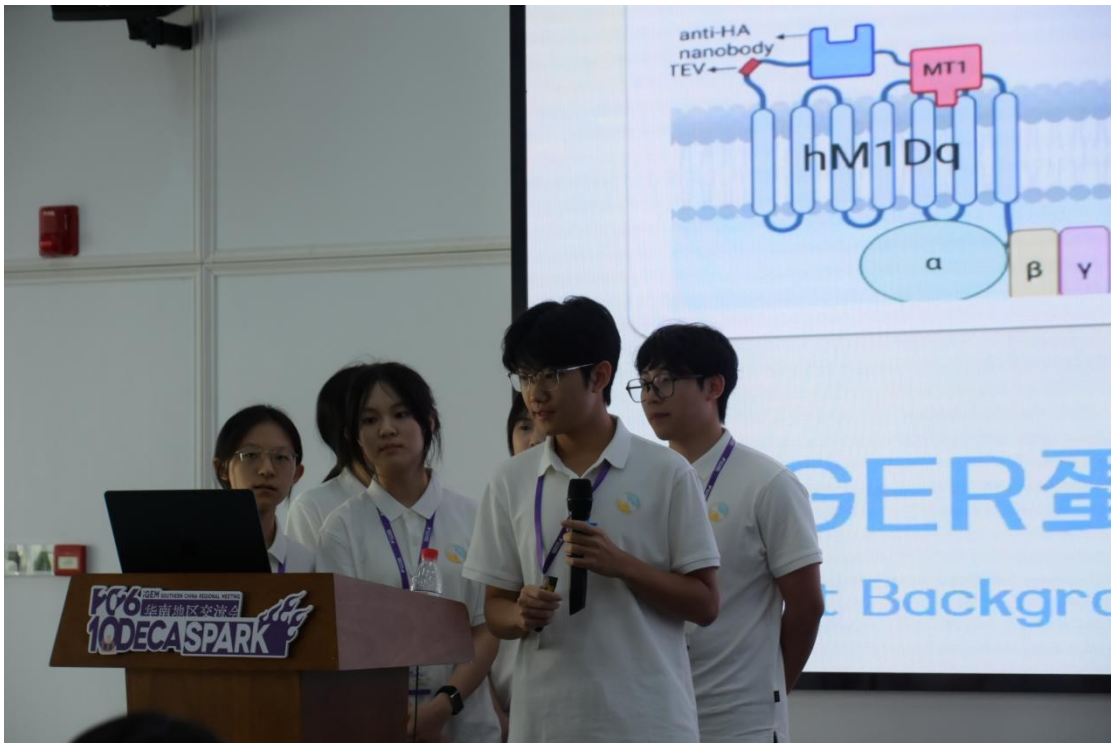


Figure 24. Photo of the team's presentation



Figure 25. Team group photo



Figure 26. Poster exchange

APiC: The First Asia-Pacific iGEM Team Exchange Meeting

APiC (Asia-Pacific iGEM Community Conference) is the first Asia-Pacific iGEM team exchange meeting, co-hosted by the Southern University of Science and Technology and the University of Hong Kong. Our team took part in both stages of the exchange, at SUSTech and at HKU, learning from and jointly refining projects with several iGEM teams from the Asia-Pacific region through lectures, mock review, project presentations and interactive activities.

SUSTech stage: mock review and project mutual visits

At SUSTech, we first took part in several expert lectures covering cutting-edge advances in synthetic biology, project validation methods and standardized practice. Afterwards, iGEM competition staff introduced the competition process and review points online, giving us a clearer understanding of this year's rules and scoring priorities.

The mock review was the highlight of this exchange. We went through the complete formal defense process, reviewing item by item from problem definition and technical route to validation design. The reviewing teachers advised us to focus further on application scenarios, add data evidence, and pay attention to compliant expression. We recorded these suggestions one by one and marked them as a list for the next round of improvement.

In the project mutual-visit session, we introduced the origin, core mechanism and expected application of the influenza A air detection project, and also learned about other teams' technical paths and validation approaches. We also studied how each team designed and organized its poster, from layout and figures to the storyline, and gained a good deal of presentation experience.

Interactive activities: making the project playable, tangible and open to feedback

We applied to the conference for an on-site interactive activity and were given dedicated space to set up two components. The first was question cards designed around the target awards: other teams could look at our questions according to their own target award and write down their answers, while we explained on the spot the design thinking behind each card and the review points of concern, and collected cross-team feedback. The second was a small "airplane battle"-style game based on the project's design concept, in which participants advanced through levels following a story guide, learning about our project in a relaxed atmosphere. Player feedback will also be used to optimize the story pacing and level design. In addition, during the exchange on experimental experience, we traded many practical tips on experimental design, quality control and data processing with other teams.

HKU stage: lectures and a discussion session with past participants

At the University of Hong Kong, we continued to attend expert lectures and joined a

discussion session with past iGEM participants. Everyone exchanged views on what they gained from competing, teamwork, and the difficulties and solutions encountered in advancing projects. The sharing by past participants gave us a more concrete understanding of the rhythm and priorities of iGEM. The HKU stage also included poster displays and free exchange sessions, where we had in-depth discussions with other teams about project design, validation schemes and presentation methods.

Takeaways and next steps

APiC was the first time we took part in an Asia-Pacific-level iGEM team exchange meeting, and it was also the first edition. Through the two stages at SUSTech and HKU, we not only learned from cutting-edge lectures and review experience, but also helped more teams learn about our project through the interactive activities we designed ourselves. Next, we will optimize the validation scheme and interaction design, refine the project materials, and continue to stay in contact and cooperate with the various teams, based on the review suggestions and what we gained from the exchange.



Figure 27. Photo of the team taking part in the mock review

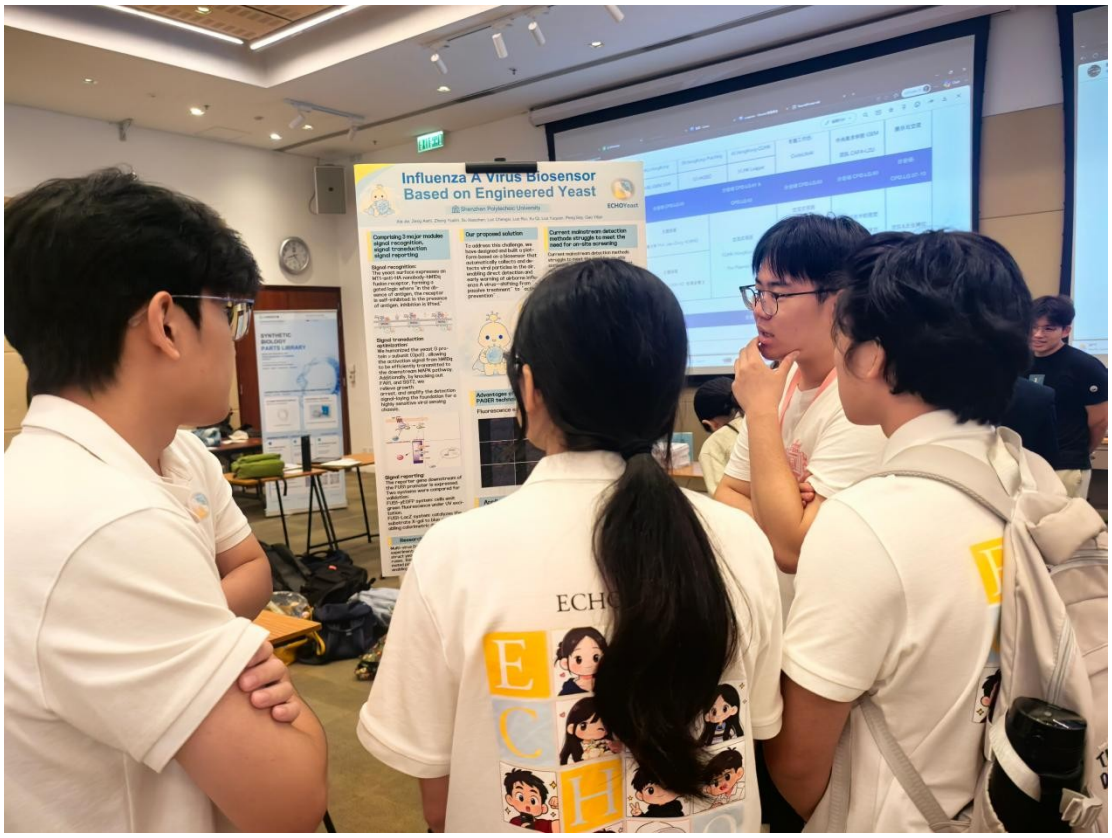


Figure 28. Photo of the team exchanging with other teams

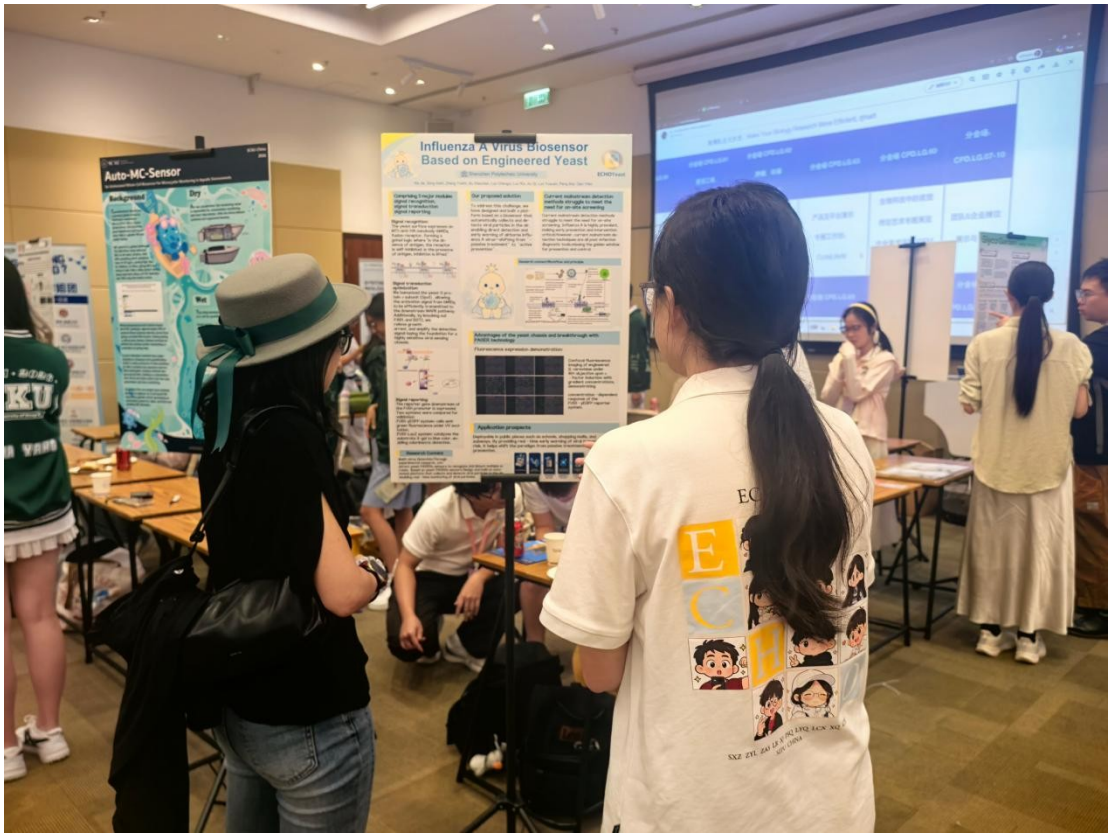


Figure 29. Photo of team members presenting the project



Figure 30. Group photo of the participating members

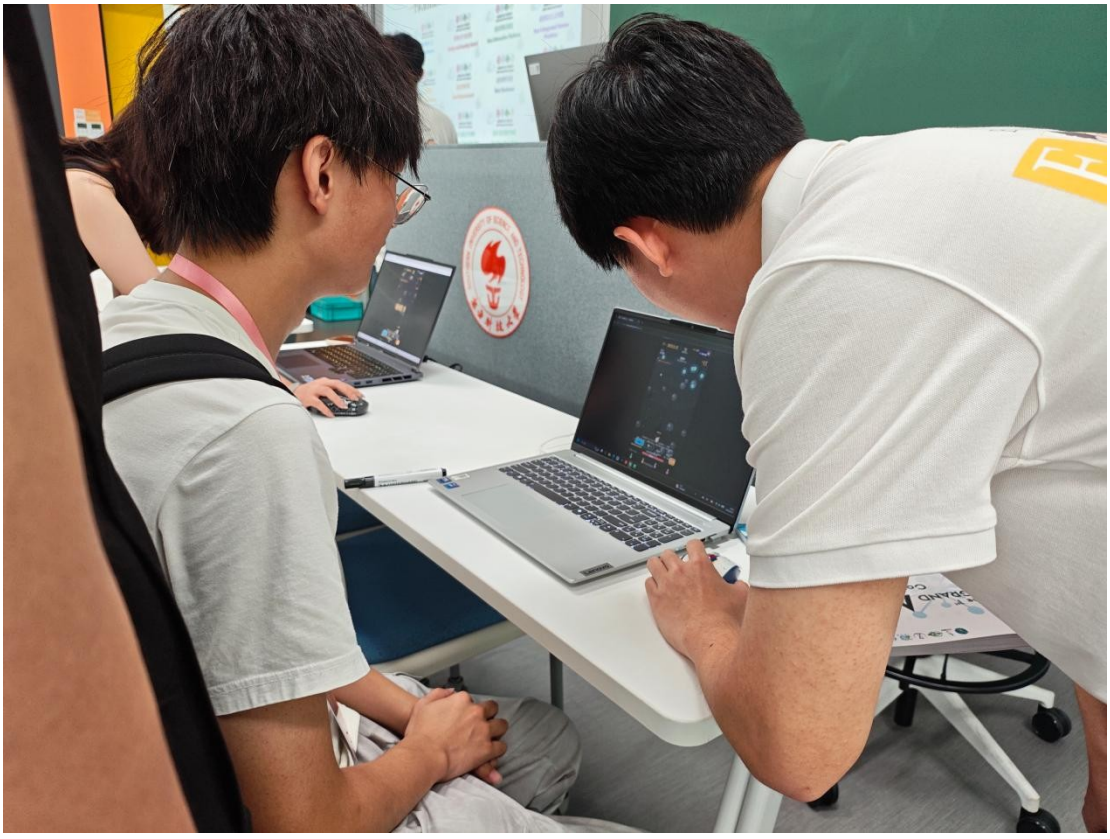


Figure 31. Photo of participation in the interactive activities

Online exchange meeting on yeast chassis engineering

Beyond APiC and the South China exchange meeting, we also took part in a themed online exchange meeting initiated by CAU-China for teams working on yeast chassis engineering. Six teams took part in total: CAU-China, NEFU-China, SZPU-CHINA, Keystone, NKU and SHSID. Unlike the first two meetings, which focused on defense rehearsal and presentation technique, this meeting went straight to the specific difficulties each team encountered in yeast genetic manipulation: each team first reported its project design and experimental progress in turn, after which free exchange began, discussing transformation efficiency, gene knockout, reporter system leakage, membrane protein expression and localization, product format and biosafety.

Technical dialogue among peers

At the meeting we gave a complete introduction to our influenza A air detection project based on Saccharomyces cerevisiae BY4741, from the antigen-gated fusion receptor and humanized G protein to the two reporter systems, yEGFP and MEL1, and honestly reported bottlenecks such as only one gene having been knocked out in the chassis and promoter leakage in the MEL1 system. We had assumed these were problems caused by our own insufficient skill in handling, but the exchange quickly revealed that low transformation efficiency, unstable auxotrophic selection, low CRISPR-Cas9 success rates and high background expression of reporter systems are difficulties faced in common by almost all yeast teams. NEFU-China shared its experience of persistently low transformation efficiency in Yarrowia lipolytica and of defective strains becoming unstable after passaging; SHSID's colorimetric module likewise got stuck at the protein expression step; and CAU-China and we had once taken the same winding road on gene knockout. This made us realize that spending a great deal of time doubting our own technique had been a misjudgment of direction, and that the root of the problem lies more in the inherent difficulty of yeast chassis engineering.

During the discussion we also contributed our own experience. CAU-China asked why we chose CRISPR-Cas9 rather than homologous recombination, and how long a single round of knockout takes; we honestly explained the one-to-two-week experimental cycle, and flagged a validation pitfall that is easily overlooked: even if the linearized fragment and the CRISPR plasmid are transformed into yeast together, the linear fragment may circularize on its own inside the cell and then complete the knockout through homologous recombination. Therefore, a sequencing result showing a frameshift mutation at the target site does not directly prove that Cas9 did the work. This reminder applies equally to other teams currently attempting this method.

Specific methods we learned

This exchange gave us a set of experimental methods that can be applied directly. CAU-China demonstrated a validation approach for membrane protein localization:

expressing the receptor as a fluorescent protein fusion, combined with FM4-64 membrane dye co-staining, so that clear plasma membrane co-localization is visible under a confocal microscope, which is precisely the link we currently lack most. Keystone built a self-destruct switch based on the mazF toxin-antitoxin system combined with an RNA thermometer: at the conventional culture temperature of 30–37 degrees Celsius the cells survive normally, but once the temperature deviates they die on their own, and this has been verified by survival experiments, providing a ready-made example for the simple, light-independent safety system we are looking for. In addition, the question-and-answer between CAU-China and NEFU-China on transformation efficiency also let us note down concrete practices: increase the concentration of the target fragment, extend the homology arms appropriately, increase the extension time and cycle number, and when there are few extraneous bands, replace gel extraction with direct recovery of the PCR product. Keystone's sodium alginate microsphere immobilization and NKU's lyophilized yeast color card gave us more optional ideas for the dosing method and product format of the engineered bacteria.

From the exchange to project improvement

After the meeting, we translated the above takeaways, item by item, into project adjustments: gene knockout was changed to a dual-track approach, continuing to optimize CRISPR-Cas9 while advancing homologous recombination as a backup route, and designing multiple parallel targets to shorten the cycle; for vector construction and transformation we adopted the practices of increasing fragment concentration, extending homology arms and directly recovering PCR products; for membrane localization validation we introduced fluorescent protein fusion combined with FM4-64 co-staining and confocal observation; and on biosafety we began investigating mazF-type light-independent suicide systems. The observation in the exchange that colorimetric output is generally blocked also provided external confirmation of the value of building the yEGFP and MEL1 reporter systems in parallel for simultaneous testing.

This exchange also showed us where we can keep improving: within the limited meeting time, we put our focus on reporting the project and answering questions, so there were correspondingly fewer opportunities to proactively consult the other teams; the presentation was delivered by a single member, which made it easier to control the overall pace and storyline, while detailed questions on each module were left more to follow-up contact after the meeting. We therefore adjusted our approach to external exchange, designating a contact person for each experimental module and sorting out the topics each module hopes to discuss before the next exchange meeting, so that the exchange can gradually move from one-way reporting toward two-way consultation.

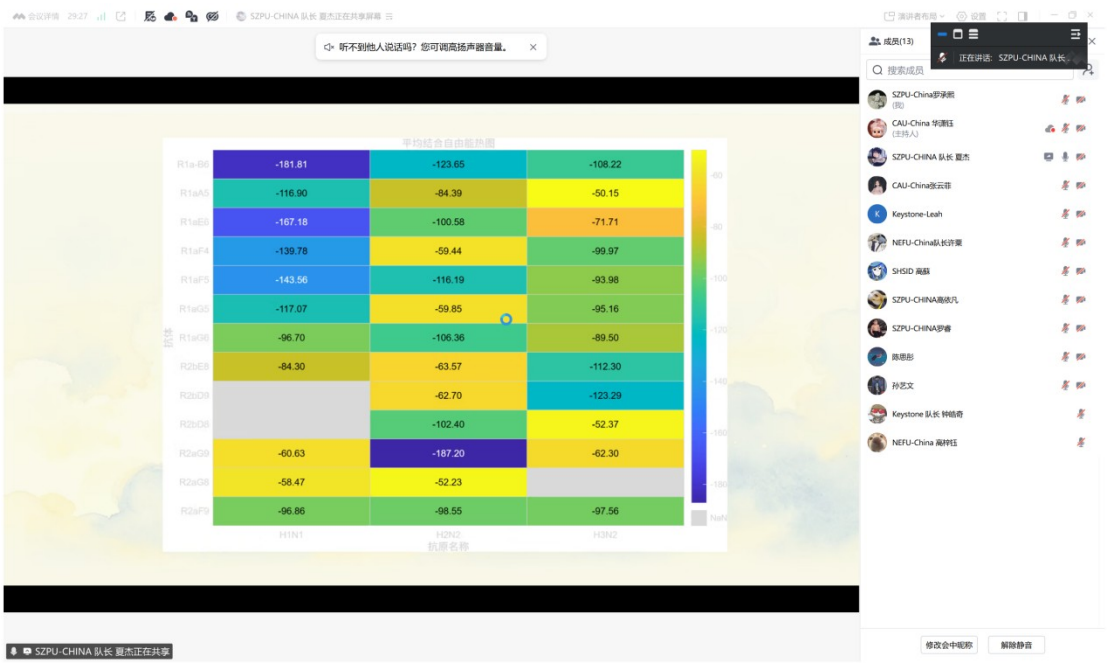


Figure 32. Photo of the exchange with other teams

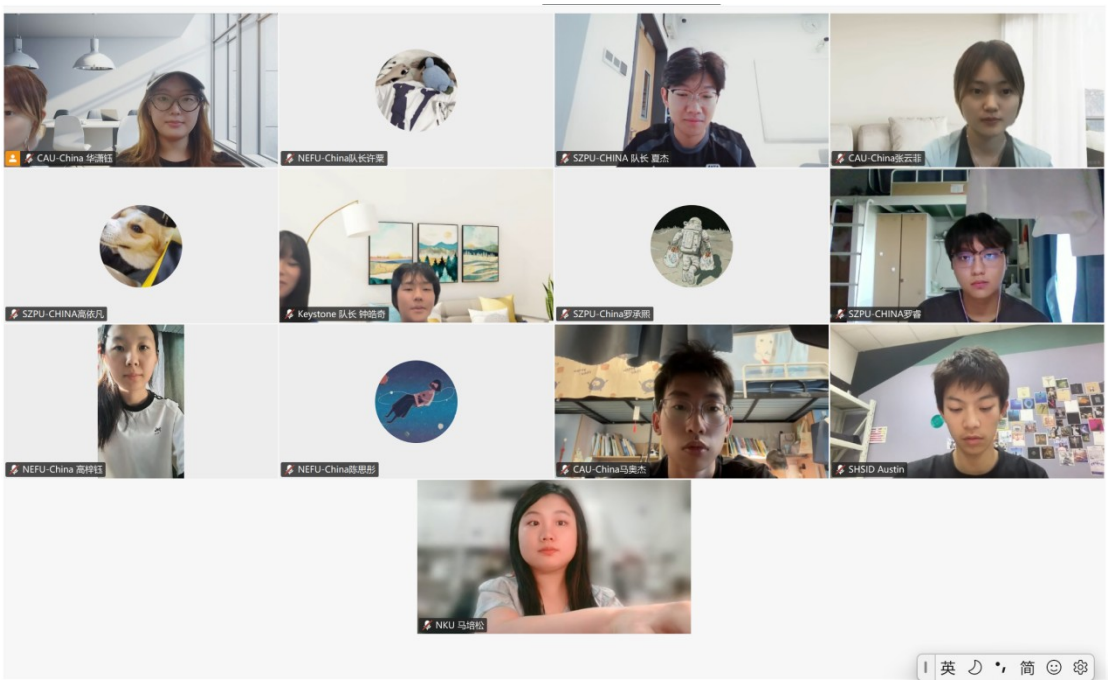


Figure 33. Group photo of the exchange meeting

Education

While advancing our research, we also hoped to bring synthetic biology out of the laboratory and let more people learn about this cutting-edge field. To this end, we gave presentations for students of Shenzhen First Vocational Technical School and for the 2026 cohort of undergraduate students in pharmacy and pharmaceutical engineering technology at our own institution. In these activities, we not only introduced the design thinking of the influenza A air detection project, but also took the opportunity to explain to the students the basic concepts, research approaches and application prospects of synthetic biology. Many of the students were encountering synthetic biology for the first time, and through our explanations they gained an intuitive understanding of "engineering microorganisms to serve human health" and developed an interest in the field. We hope that through such presentations more people can learn about and understand synthetic biology, and that the project can accumulate broader social feedback.

iGEM science outreach session for incoming students

The session was delivered jointly by the iGEM student team and the supervising teachers. We started from synthetic biology itself to help the students build a basic understanding: it is not merely gene editing, but requires the engineering mindset of "design-build-test-learn" to solve real-world problems in healthcare, environmental governance and new materials development. Using everyday examples, we broke down the stereotype that "synthetic biology is far removed from daily life," letting the students appreciate that the discipline is both interesting and rigorous.

We introduced the content of research projects, academic competitions, academic conferences and social practice, emphasizing that extracurricular practice needs to be built on a solid foundation in major courses. Given that most of the students had never encountered iGEM, we introduced this global student competition in synthetic biology: it is not limited to wet experiments but also covers hardware construction, modeling, human practices, wiki building, poster defense and other modules, and can train comprehensive abilities such as literature research, experimental operations, stakeholder interviews and teamwork.

We focused on introducing the airborne influenza A virus biosensor project, explaining its positioning as an early warning of infection risk in enclosed spaces, and shared our interview experience with experts in disease control, health administration, hardware development and synthetic biology, showing how expert opinions were

genuinely applied to project iteration. During the activity we handed out a popular-science handbook on influenza A protection written by the team ourselves, linking laboratory research with public health needs; the supervising teachers also added their interpretation of the relationship between major course study and science and innovation practice.

In terms of outcomes, most participants established a basic understanding of synthetic biology, the iGEM competition and the influenza A detection sensor project, and also gained a preliminary understanding of content related to genetic engineering, dispelling the fixed impression that "research is very remote from young learners." The activity also delivered popular-science education on influenza A protection, raising the students' awareness of protection against respiratory infectious diseases. The shortcoming is that, limited by the presentation time, we could not explain the experimental principles and hardware details in depth; and the handbook was distributed in bulk without item-by-item explanation on site. Going forward we plan to open consultation channels for interested participants, hold small themed salons, and collect handbook feedback to iterate subsequent outreach materials.



Figure 34. Photo of the supervising teacher introducing the team


Figure 35. Photo of team members giving the presentation

Synthetic biology outreach at a secondary vocational school

We gave a presentation for students in pharmacy-related majors at Shenzhen First Vocational Technical School, with team members doing the sharing. These students' daily study leans toward traditional pharmaceutical processes, they rarely have the chance to encounter cutting-edge content in synthetic biology, and they easily form a "profound and hard to understand" stereotype about life science research.

Here too we prioritized outreach about synthetic biology itself, and combining it with the current state of the pharmaceutical industry, we explained the application prospects of synthetic biology in directions such as drug synthesis and biological detection, letting the students see a cutting-edge track that their own major can extend into. Through accessible examples, we showed that synthetic biology is both fun and creative and scientifically rigorous, dispelling the students' fear of cutting-edge research and letting them know that young students can equally take part in hands-on synthetic biology innovation projects.

We introduced the iGEM competition and the value of science and innovation practice for young people, explaining that the competition is not just laboratory experiments but a science and innovation activity combining experiments, hardware, design and social research across multiple dimensions. In the project introduction, we used the airborne influenza A virus biosensor as an example, explaining the project's inspiration, background, experimental design, hardware design, wiki page and IP visual design; team members with a pharmacy background also shared their experience of self-teaching programming out of interest, showing that interdisciplinary skills are not something only those in related majors can master.

We also shared the team's human practices, academic exchanges and competition experience, and handed out custom merchandise to increase on-site engagement. In the interactive session, we answered the students' questions while collecting genuine suggestions from end users, such as improving the device's portability and shortening detection time.

For the participating students, this sharing let them get close to research content related to synthetic biology, see in advance the research and industry development directions corresponding to their own major, step outside the confines of the textbook, and deepen their three-dimensional understanding of their field. They came to realize that research is not only test-tube experiments but also includes programming, design and science communication; and through a public-health-oriented project case, they also appreciated the value of technology in solving real social problems, with some students beginning to think about the connection between their personal interests and future major choices. For the iGEM team, in order to explain the project clearly to a non-specialist audience, we reorganized and simplified the complex technical logic, deepening all members' understanding of the project as a whole; the suggestions raised by participants, such as portability and detection time, also became important references for prototype iteration and refinement of application scenarios. The shortcoming is that time was limited, so a deep hands-on workshop could not be

carried out. In the future we hope to build a regular channel for science outreach and exchange aimed at young people.



Figure 36. Photo of the team introducing the iGEM competition



Figure 37. Group photo of the outreach presentation

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